

FPT UNIVERSITY

Capstone Project Document

Lightweight Deep Learning for Robust Shrimp Disease
Detection on Mobile Devices under Diverse Environmental
Conditions

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Capstone Project code	AIP491_01

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Acknowledgement

The team sincerely thanks Nguyen Dinh Vinh, Lecturer and Supervisor, for his academic guidance, milestone reviews, and detailed feedback throughout the capstone project. We acknowledge FPT University Can Tho for the academic environment and project framework. We also acknowledge the authors and maintainers of the public shrimp-image datasets, open-source frameworks, and research tools used in the classification, instance-segmentation, robustness, explainability, and mobile evaluation work reported here. Finally, the team thanks the lecturers, colleagues, friends, and families who supported experimentation, application development, documentation, presentation, and defense preparation.

Definition and Acronyms

The following terms are used throughout this standalone report. The list is limited to terminology that appears in Report 6.

Acronym	Definition
AP	Average Precision
ASL	Asymmetric Loss
ASL-LDAM	Custom single-label objective combining ASL-style asymmetric focusing with Label-Distribution-Aware Margin
BG	Black Gill disease
CA	Coordinate Attention
CE	Cross-Entropy
COCO	Common Objects in Context evaluation convention used by the retained MMDetection comparator
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
DCFR	Project-created texture feature-recalibration component used with SimAM
DPCA	Dual-Polarity Contrast Attention
EXT-3-Original	Project identifier for the three-class external ShrimpDB retraining source
FP16	16-bit floating-point model precision
FP32	32-bit floating-point model precision
FPS	Frames Per Second
GFLOPs	Billions of floating-point operations
GPU	Graphics Processing Unit
Grad-CAM++	Generalized gradient-based class activation mapping
HiResCAM	High-Resolution Class Activation Mapping
HScore	Project-defined healthy-aware instance-segmentation decision score
HSV	Hue, Saturation, and Value color representation
IoU	Intersection over Union
Kappa	Cohen's agreement statistic reported on a 0-100 display scale where stated
LKA	Large Kernel Attention
MAE	Mean Absolute Error
mAP50	Mean Average Precision at an IoU threshold of 0.50
mAP50-95	Mean Average Precision over IoU thresholds 0.50 to 0.95
NMS	Non-Maximum Suppression
RGB	Red, Green, and Blue color representation
RTMDet	Real-Time Models for object detection; retained here as an external instance-segmentation comparator family
SDI-4	Primary four-class ShrimpDiseaseImageBD-4 project dataset regime
SimAM	Simple, parameter-free attention module based on neuron energy
TFLite	TensorFlow Lite
TIMM	PyTorch Image Models library
WSSV	White Spot Syndrome Virus
WSSV_BG	Four-class image-level co-infection category containing BG and WSSV evidence
XAI	Explainable Artificial Intelligence
YOLO	You Only Look Once model family

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VI. Experimental Evaluation, Results, and Discussion

1. Experimental Evidence Overview

This chapter reports the verified outcomes of the two computer-vision tasks defined in Chapter IV. The classification branch evaluated a four-class, single-label problem in the fixed class order Healthy, BG, WSSV, and WSSV_BG. The retained evidence covers candidate-model screening, the clean comparison between the YOLO26m-cls cross-entropy (CE) reference and the final ASL-LDAM + SimAM-DCFR configuration, controlled synthetic corruption tests, additional-source experiments, qualitative explainability, and a bounded Android deployment observation. The instance-segmentation branch evaluated BG and WSSV disease-region masks under specimen-aware protocols, with separate analyses of split leakage, baseline families, attention configurations, Fourier preprocessing, corruption sensitivity, and Healthy negative-image behavior.

Machine-readable values preserved in the final evidence were used directly where available. Unsupported row-level values were not reconstructed from memory or estimated from plots. Results are reported on a 0–100 percentage scale unless otherwise stated, and absolute changes are expressed in percentage points.

2. Dataset Composition and Evaluation Partitions

2.1 Classification Dataset Composition

The primary classification source is SDI-4, the 1,149-image ShrimpDiseaseImageBD-4 primary dataset containing Healthy, BG, WSSV, and WSSV_BG. EXT-3-Original denotes the separately mapped external ShrimpDB source with three trained classes: Healthy, BG, and WSSV. SDI-4 + EXT-3-Original denotes the four-class partition-wise integrated regime created by combining corresponding source-specific training, validation, and test partitions; WSSV_BG is contributed only by SDI-4.

Table 6.1: Classification dataset regimes and fixed partition counts

Dataset regime	Classes	Train	Validation	Test	Total
SDI-4	4	804	172	173	1,149
EXT-3-Original	3	221	47	47	315
SDI-4+EXT-3-Original	4	1,025	219	220	1,464

Assessment. SDI-4 remains the primary four-class regime; EXT-3-Original is a separate three-class re-training source, and the integrated regime is a larger four-class retraining experiment rather than direct unseen-source inference.

Each source was split independently before partition-wise integration. The retained multi-source results are retraining experiments on the stated dataset regimes, not direct inference by the original SDI-4 checkpoint on an unseen source.

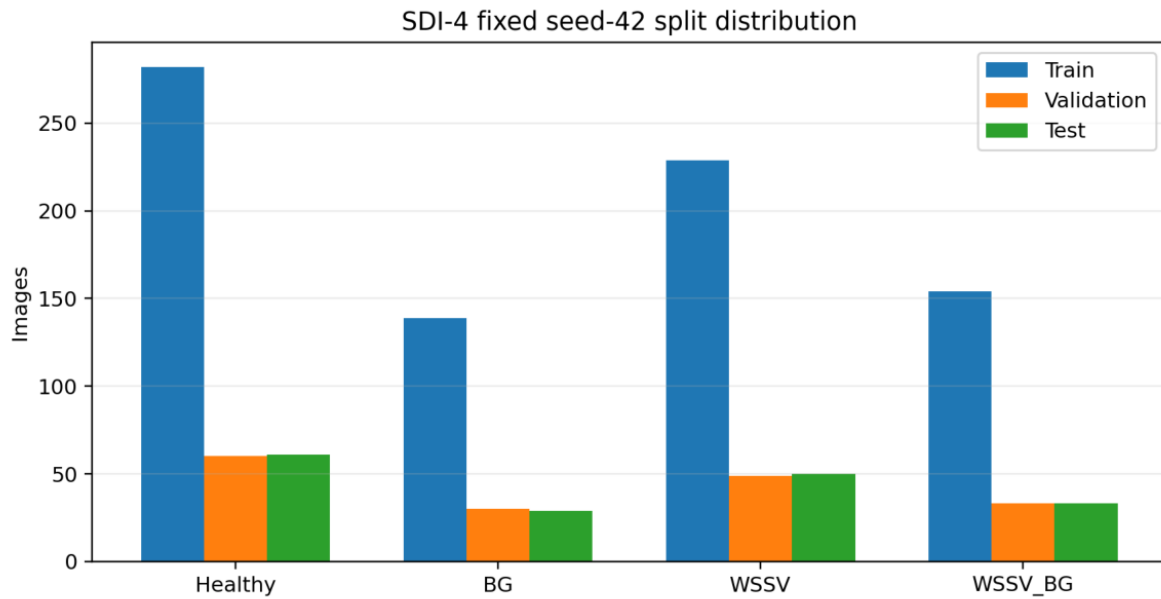


Fig. 6.1. Class distribution across the fixed seed-42 SDI-4 training, validation, and test partitions.

2.2 Instance-Segmentation Dataset Composition

The recognized specimen-group subset contained 1,149 images associated with 416 filename-derived specimens. Project-authored BG or WSSV disease-region labels were available for 746 images, while 403 Healthy images were retained as empty-label negatives containing no positive disease mask. Healthy is therefore an image-status condition used during training and evaluation, not a third segmentation-mask class.

Table 6.2A: Recognized specimen-group subset under the 80/10/10 protocol

Split	Images	Specimens	Labeled disease images	Healthy empty-label images
Train	905	331	584	321
Validation	115	40	74	41
Test	129	45	88	41
Total	1,149	416	746	403

Assessment. The filename-compatible subset supports specimen-grouped evaluation across 1,149 images and 416 recoverable specimen groups, while retaining 403 Healthy empty-label negatives for false-positive analysis.

The expanded image inventory retained all 1,149 recognized specimen-group images and added 303 unmatched images. The added images were allocated by the retained project logic because compatible specimen identifiers were unavailable.

Table 6.2B: Expanded instance-segmentation image inventory

Split	Images	Recognized specimen-group images	Added unmatched images
Train	1,147	905	242
Validation	144	115	29
Test	161	129	32
Total	1,452	1,149	303

Assessment. The 303 unmatched-name images expand disease coverage but cannot be assigned by specimen identity from filenames; the 1,452-image inventory therefore has a mixed grouped and image-level allocation boundary.

2.3 External and Partition-Wise Integrated Classification Datasets

EXT-3-Original maps the source folders Tom_BT, Den_Mang, and Dom_Trang to Healthy, BG, and WSSV, respectively. Its fixed partition contains 221 training images, 47 validation images, and 47 test images. The suffix “-3” identifies the three trained classes and distinguishes this project-defined experiment identifier from the official source name ShrimpDB.

SDI-4 + EXT-3-Original is the four-class partition-wise integrated regime. SDI-4 and the external source were split independently, after which train was integrated with train, validation with validation, and test with test. The resulting partition contains 1,025 training images, 219 validation images, and 220 test images. WSSV_BG remains available only from SDI-4. Because source composition, class priors, and acquisition backgrounds change, these results qualify the primary finding rather than establishing universal generalization.

3. Classification Baseline Screening

3.1 ShrimpXNet-Oriented Reference Context

ShrimpXNet was retained as the closest literature-oriented classification reference because its published pipeline is conceptually close to two design choices used in this project: foreground-focused image preparation and ConvNeXt-Tiny-style transfer learning. Its role in this chapter is therefore contextual rather than architectural. The final CVio classifier is not a modified ShrimpXNet, and no result in this report should be read as a direct reimplement or extension of that model. The reference is used to explain why background suppression and transferable visual features were reasonable starting points before the project performed its own candidate screening.

To keep the subsequent comparisons internally consistent, the team did not import ShrimpXNet results into the project ranking table. Candidate models were instead evaluated under the project-owned fixed seed-42 classification partition, with cross-entropy serving as the controlled objective during baseline screening and reference-model training. This separation matters because a literature result obtained under a different dataset composition, preprocessing pipeline, split policy, or training budget cannot be treated as numerically interchangeable with the retained SDI-4 evidence. ShrimpXNet consequently provides a research orientation, while the model-selection evidence is generated from the project's own controlled records.

3.1 ShrimpXNet-Oriented Reference Context (continued)

The broader screening strategy also extends beyond the architectural scope suggested by the literature reference. Rather than assuming that one ConvNeXt-style backbone was automatically the best mobile candidate, the project examined a heterogeneous set of TIMM and Ultralytics classifiers and considered predictive quality together with parameter count, model size, export feasibility, and retained timing metadata. This makes the reference useful without allowing it to predetermine the final architecture. It also avoids attributing later gains from ASL-LDAM or SimAM-DCFR to ShrimpXNet, because those components were integrated only after the project had selected its own YOLO26m-cls reference.

Accordingly, the interpretation boundary for this subsection is deliberately narrow. ShrimpXNet supports the motivation for foreground-aware, transfer-learning-based shrimp classification, but it is not used as a head-to-head numerical baseline under the seed-42 protocol. The quantitative baseline begins with the project screening described in Sections 3.2 and 3.3. This distinction keeps literature context, candidate selection, and the final proposed configuration as separate evidence layers instead of collapsing them into one model lineage.

3.2 Complete TIMM Candidate Benchmark

The revised lightweight selection preference is a deployment-oriented budget of approximately 15 million parameters or fewer. Historical screening artifacts also contain larger diagnostic references; those rows are retained only when they help explain capacity behavior and are not treated as eligible final mobile candidates. This distinction is important for the YOLO26m-cls versus YOLO26x-cls comparison: YOLO26x-cls is useful as a higher-capacity diagnostic control, but its 28.36M parameters place it outside the revised deployment preference.

The retained TIMM screening and Ultralytics screening used the same fixed seed-42 classification partition. Within the deployable range, the objective was not to minimize parameter count alone, but to find a stable predictive-efficiency trade-off using Macro-F1 as the principal outcome together with accuracy, model complexity, export feasibility, and measured runtime metadata.

The complete retained seed-42 TIMM screening ledger is reported in Tables 6.3A and 6.3B on the following pages. Predictive metrics and model complexity are separated from timing metadata so that all 17 retained candidates remain readable at the thesis table font size. The uploaded CSV was cross-checked against the retained screening image before typesetting; no missing TIMM row required numerical imputation.

Table 6.3A: Complete TIMM seed-42 predictive and complexity screening results

Model	Params (M)	Val F1 (%)	Test Accuracy (%)	Test Macro-F1 (%)
convnext_tiny_in22k	27.82	85.30	85.22	84.16
mobilenet_v3_large	4.21	79.27	82.61	80.10
efficientnet_b0	4.01	83.20	79.13	77.01
repvgg_a0	7.83	78.49	77.39	76.88
efficientnet_v2_s	20.18	78.27	77.39	76.78
shufflenet_v2_x1_0	1.26	76.17	74.78	71.85
fastvit_t8	3.26	78.29	73.91	71.03
edgenext_xx_small	1.16	71.46	72.17	70.63
mobileone_s0	4.27	73.34	71.30	69.69
mobilevit_s	4.94	80.78	73.91	69.18
mobilenetv4_conv_small	2.50	75.65	67.83	66.26
mnasnet_100	3.11	70.79	66.96	62.48
ghostnetv2_100	4.88	72.40	61.74	61.01
rexnet_100	3.52	63.99	61.74	58.55
squeezenet1_1	0.72	68.86	59.13	55.04
mobilenetv4_hybrid_medium	9.80	59.43	60.87	53.83
efficientvit_m1	2.79	68.10	57.39	52.83

Assessment. ConvNeXt-Tiny-IN22K is the strongest retained TIMM row by test Macro-F1 but exceeds the approximate 15M-parameter preference; MobileNetV3-Large is the strongest retained TIMM row within that preference.

ConvNeXt-Tiny-IN22K produced the strongest retained TIMM test Macro-F1 (84.16%) and test accuracy (85.22%), but its 27.82M parameters place it outside the project's approximately 15M-parameter deployment preference. Within that preference, MobileNetV3-Large was the strongest retained TIMM row by test Macro-F1 (80.10%) at 4.21M parameters. These results show that the TIMM screen supplied useful lightweight references, but none matched the later YOLO26m-cls reference on the locked primary screening metric.

Table 6.3B: Complete TIMM seed-42 training and inference timing ledger

Model	Training Time (s)	Reported Inference Time (s)	FPS	Latency (ms)
convnext_tiny_in22k	152.0	2.34	49.2	20.34
mobilenet_v3_large	234.1	2.52	45.6	21.93
efficientnet_b0	235.3	2.47	46.5	21.52

repvgg_a0	129.3	2.39	48.1	20.81
efficientnet_v2_s	195.4	3.57	32.2	31.06
shufflenet_v2_x1_0	226.0	2.51	45.7	21.86
fastvit_t8	110.1	2.56	44.8	22.30
edgenext_xx_small	181.3	2.39	48.2	20.75
mobileone_s0	136.6	4.11	28.0	35.76
mobilevit_s	150.7	2.47	46.5	21.51
mobilenetv4_conv_small	178.1	2.38	48.2	20.74
mnasnet_100	180.4	2.39	48.2	20.76
ghostnetv2_100	92.3	3.30	34.8	28.71
rexnet_100	129.2	2.54	45.3	22.06
squeezenet1_1	135.5	2.26	50.8	19.67
mobilenetv4_hybrid_medium	147.8	3.38	34.0	29.43
efficientvit_m1	148.9	3.17	36.3	27.55

Assessment. SqueezeNet1_1 is the fastest retained TIMM candidate, but its low test Macro-F1 shows why throughput cannot be used as a stand-alone selection criterion.

The timing ledger also demonstrates why speed was not used as a stand-alone selection rule. SqueezeNet1_1 recorded the highest retained TIMM throughput (50.8 FPS; 19.67 ms), but its test Macro-F1 was only 55.04%. Conversely, several stronger predictive models remained near the 20-22 ms range. The screening decision therefore considered predictive quality and efficiency jointly.

3.3 Ultralytics Classification Benchmark

The Ultralytics benchmark formed the second classification screening stage after the broader TIMM survey. Its purpose was not to claim that one YOLO family is universally superior, but to identify a practical reference architecture that could support the later controlled CE comparison and remain plausible for mobile deployment. The retained search covered classification candidates from the YOLOv8, YOLO11, and YOLO26 families under the same fixed seed-42 SDI-4 partition used for the project screening. Cross-entropy remained the reference objective at this stage so that architecture choice was separated from the later loss-function and feature-recalibration modifications.

Although the historical inventory contained more Ultralytics variants, Table 6.3 reports five representative rows for which predictive and efficiency metadata were jointly traceable in the retained evidence. The table is therefore a selection summary rather than a complete family leaderboard. Macro-F1 was treated as the principal predictive signal because the four-class problem is not adequately described by overall accuracy alone, while accuracy, parameter count, serialized size, and the retained batch-one GPU latency/FPS measurements were used as supporting engineering evidence. The project-specific preference of approximately 15 million parameters was a deployment-oriented design target, not a theoretical cutoff and not a claim that models above that size are intrinsically unsuitable.

3.3 Ultralytics Classification Benchmark (continued)

The representative rows expose several useful trade-offs. YOLOv8s-cls is substantially smaller and faster in the auxiliary screening record, but its Macro-F1 is lower than the selected YOLO26m-cls. YOLO26x-cls provides a deliberately larger capacity point: it matches the 89.02% accuracy of YOLO26m-cls but records lower Macro-F1 while using 28.36M parameters and a 54.37 MB artifact in the retained summary. YOLO11m-cls has a footprint close to YOLO26m-cls but lower retained predictive scores. These comparisons show why the decision cannot be reduced to either maximum throughput or maximum capacity.

Timing provenance is kept separate from predictive selection. The latency and FPS values in Table 6.3 are auxiliary GPU screening measurements associated with the retained classification checkpoints; they are not Android application latency, and they are not measurements from the final dual-T4 training protocol. They should therefore be used only to describe the relative efficiency evidence preserved for this screening stage. Cross-device Android runtime is reported later in Section 14 under a different execution contract that includes application-side preprocessing, conditional dispatch, model execution, postprocessing, and other runtime overhead.

Within these boundaries, YOLO26m-cls emerged as the strongest practically sized representative candidate on the locked screening signal, reaching 89.02% Macro-F1 with 10.36M parameters in Table 6.3. The result justifies carrying this architecture forward as the controlled reference, but it does not establish statistical significance, multi-seed stability, or superiority over every untested classifier. Those limitations are retained explicitly because the historical screening procedure used one fixed split and inspected test-partition Macro-F1 during model-family selection.

3.4 Reference-Model Selection

Reference-model selection was treated as a multi-objective engineering decision rather than a single-metric ranking. Predictive performance had priority, but the candidate also had to remain compatible with the project's lightweight deployment direction. In the representative Ultralytics summary, YOLO26m-cls combines 89.02% accuracy and 89.02% Macro-F1 with a 10.36M-parameter footprint and a retained 19.92 MB screening artifact. YOLO26x-cls does not provide a compensating predictive advantage: it matches accuracy, falls to 88.13% Macro-F1, and increases the footprint to 28.36M parameters and 54.37 MB. Conversely, YOLOv8s-cls is smaller and faster in the auxiliary timing record but gives up Macro-F1. The selected reference therefore sits on a more favorable predictive-efficiency compromise for the stated project preference.

The separate YOLO26m-cls versus YOLO26x-cls train/validation diagnostic provides an additional capacity check without being treated as a new test experiment. Under the same seed-42 CE protocol, YOLO26m-cls reached a best validation Top-1 accuracy of 84.30% and finished at 82.56%, whereas YOLO26x-cls reached and finished at 80.23%. Final validation loss was also lower for YOLO26m-cls, 0.4134 versus 0.4749. These trajectories do not prove that the larger model uniquely overfit, but they show that additional capacity did not improve the retained validation trajectory or the historical test Macro-F1. After this decision, YOLO26m-cls was carried forward as the fixed architecture for the controlled CE baseline and the later ASL-LDAM + SimAM-DCFR configuration, which keeps backbone selection separate from the proposed modifications.

The selection evidence still has a known validity boundary. The historical workflow used test-partition Macro-F1 as a model-family selection signal, so repeated inspection of that partition can bias the apparent ranking. This chapter preserves that protocol rather than relabeling it as validation-only selection, and it does not claim confidence intervals or multi-seed significance for the screening differences. A stronger future protocol would perform architecture and hyperparameter selection on validation data only, keep an untouched final test partition, and repeat the principal comparison across several seeds. Within the evidence actually retained, YOLO26m-cls remains the most defensible reference for the subsequent controlled experiments.

Table 6.3: Verified representative Ultralytics selection summary

Model	Accuracy (%)	Macro-F1 (%)	Params (M)	Size (MB)	Latency (ms)	FPS
YOLO26m-cls	89.02	89.02	10.36	19.92	3.16	316
YOLO26x-cls	89.02	88.13	28.36	54.37	4.59	218
YOLOv8m-cls	87.86	87.18	15.78	30.22	2.44	410
YOLOv8s-cls	87.86	87.10	5.09	9.79	1.87	535
YOLO11m-cls	87.28	86.75	10.36	19.92	3.16	317

Assessment. YOLO26m-cls provides the strongest retained Macro-F1 among the practically sized representative candidates; YOLO26x-cls is larger, while YOLOv8s-cls is faster and smaller but less accurate by Macro-F1.

Within this representative set, YOLO26m-cls produced the highest Macro-F1 while remaining below the approximate 15-million-parameter preference. YOLO26x-cls matched its accuracy but had lower Macro-F1 and substantially greater size. YOLOv8s-cls was faster and smaller in the auxiliary screening measurement, but its Macro-F1 was lower. These trade-offs support the engineering selection of YOLO26m-cls within the evaluated candidate set; they do not establish universal superiority over untested architectures.

A separate train/validation diagnostic was retained specifically to test whether the additional capacity of YOLO26x-cls improved optimization or generalization under the same seed-42 protocol. No new test inference was performed for this diagnostic. YOLO26m-cls reached a best validation Top-1 accuracy of 84.30% at epoch 25 and finished at 82.56%, whereas YOLO26x-cls reached 80.23% at epoch 29 and finished at the same 80.23%. Final validation loss was also lower for YOLO26m-cls (0.4134) than for YOLO26x-cls (0.4749). Both runs show some train-validation separation, but the evidence does not justify claiming that YOLO26x alone “overfit.” The defensible conclusion is narrower: extra capacity did not improve validation performance or the historical test Macro-F1 under the fixed protocol.

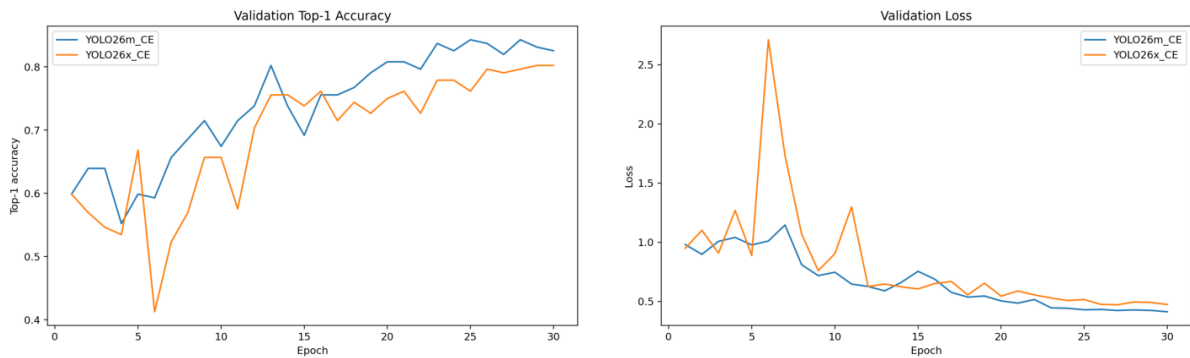


Fig. 6.1A. YOLO26m-cls versus YOLO26x-cls validation diagnostics under the same seed-42 CE protocol. Left: validation Top-1 accuracy by epoch. Right: validation loss by epoch. The larger YOLO26x-cls model did not improve the validation trajectory and is outside the revised approximately 15M-parameter deployment preference. These curves are diagnostic training/validation evidence and do not replace the locked primary clean-test result.

The historical screen used test-partition Macro-F1 as a selection signal. Repeated inspection of that partition can bias model-family ranking. This chapter preserves the actual protocol rather than relabeling it as validation-only selection, and it does not claim multi-seed significance.

4. Main Classification Performance

4.1 CE Baseline versus Final Proposed Model

The primary comparison used the same YOLO26m-cls reference architecture, fixed seed-42 split, background-removed representation, optimizer family, schedule, epoch budget, and evaluation definitions. Cross-entropy defined the reference condition. The final configuration replaced the training objective with ASL-LDAM and inserted SimAM-DCFR before the classification head.

Table 6.4: Main clean-test classification comparison

Method	Accuracy (%)	Macro-F1 (%)	Kappa $\times 100$	Parameters	Δ Macro-F1 (pp)
YOLO26m-cls + CE	89.02	89.02	85.05	10.36M	0.00
YOLO26m-cls + ASL-LDAM + SimAM-DCFR	91.33	91.01	88.16	10,886,467	+1.99

Assessment. The final ASL-LDAM + SimAM-DCFR configuration improves accuracy, Macro-F1, and Kappa over the CE reference, including a +1.99 percentage-point Macro-F1 gain under the recorded fixed seed-42 protocol.

The final model increased Macro-F1 by 1.99 percentage points and accuracy by 2.31 points. Cohen's Kappa increased from 0.8505 to 0.881593, corresponding to approximately +3.11 points on the displayed $\times 100$ scale. Under the recorded fixed-seed protocol, the improvement indicates a better balance of classwise predictions, but it is not a confidence interval or a statistical-significance result.

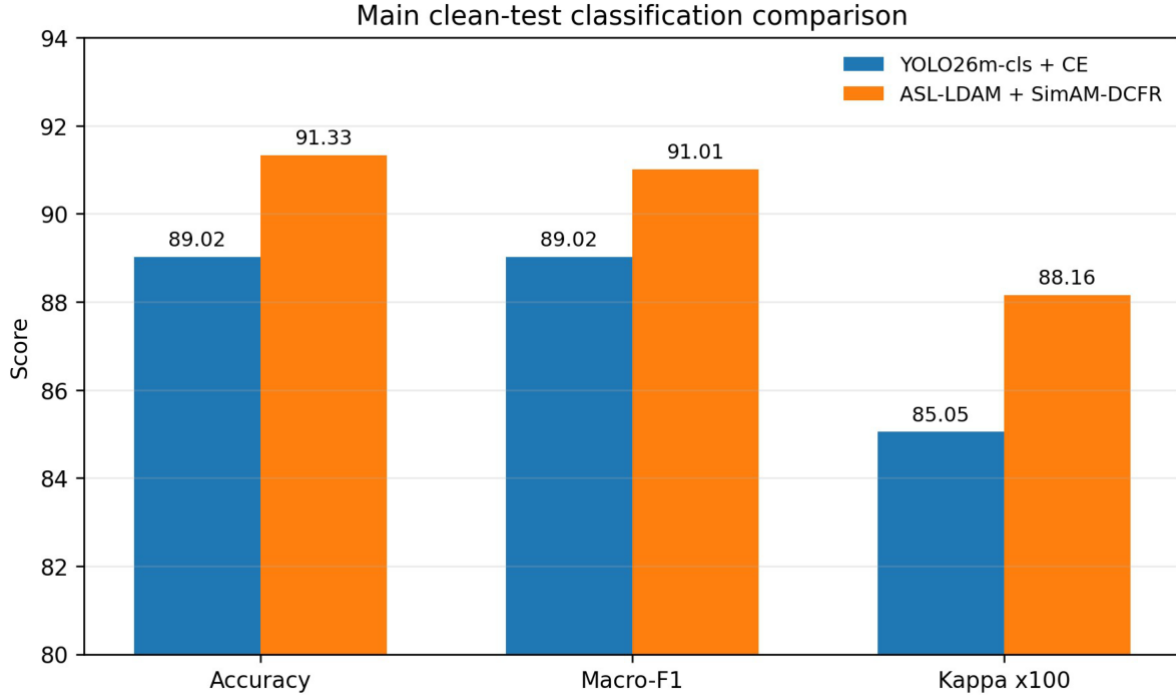


Fig. 6.2. Accuracy, Macro-F1, and Cohen's Kappa for the clean CE reference and final proposed classifier.

4.2 Classwise Performance

The final clean-test comparison establishes an aggregate improvement, but the retained primary artifacts are insufficient for a defensible class-by-class decomposition of that gain. The 173-image SDI-4 primary test partition does not have a preserved prediction-level ledger or complete classification report for the final checkpoint. Exact primary-test precision, recall, F1, support, confidence values, and class-transition counts are therefore not reconstructed. This is more than a documentation inconvenience: Macro-F1 is an average over classes, so the increase from 89.02% to 91.01% does not reveal whether every class improved, whether one difficult class improved substantially while another regressed, or whether precision and recall moved in opposite directions for a specific disease category.

Reconstructing per-class values from the reported aggregate metrics would be methodologically unsafe. Accuracy constrains only the total number of correct predictions, while Macro-F1 summarizes non-linear functions of classwise precision and recall. Many different confusion structures can produce similar aggregate values. Support counts alone are also insufficient because they do not identify which samples moved between Healthy, BG, WSSV, and WSSV_BG. For that reason, this report treats the missing primary classwise ledger as a material evidence limitation rather than estimating plausible numbers from rounded metrics or figures.

Section 8 provides genuine classwise evidence for the separately trained SDI-4 + EXT-3-Original regime, including the observed high-precision/lower-recall WSSV pattern and the higher-recall/lower-precision WSSV_BG pattern. Those results are useful for discussing a plausible disease-versus-co-infection trade-off, but they cannot be substituted for the primary 173-image experiment. The integrated regime contains 220 test images, changes source composition and class priors, and introduces an external source for Healthy, BG, and WSSV while WSSV_BG remains supplied only by SDI-4. A classwise pattern observed there may therefore reflect the altered retraining regime as well as the model.

4.2 Classwise Performance (continued)

A complete primary analysis should be regenerated from a frozen prediction file tied to the final checkpoint, exact seed-42 split manifest, and fixed class order. It should report per-class precision, recall, F1, support, confidence distributions, and paired CE-versus-proposed transitions on the same 173 images. Until that artifact exists, the defensible claim is limited to the verified aggregate improvement and the separate extension evidence. This keeps the main result useful without implying class-specific behavior that the retained primary records cannot prove.

4.3 Confusion-Matrix Analysis

The same provenance constraint applies to confusion-matrix analysis. No verified count matrix or normalized matrix was retained for the final proposed classifier on the 173-image primary SDI-4 test partition. A confusion matrix cannot be derived uniquely from accuracy, Macro-F1, Kappa, or a small set of classwise summaries because those statistics discard the exact sample-to-sample transition structure. Accordingly, this chapter does not fabricate a primary matrix or infer cell counts from rounded percentages.

The normalized matrix shown later in Section 8 belongs specifically to the SDI-4 + EXT-3-Original retraining experiment. It demonstrates why a matrix is useful by exposing class-specific confusion that is hidden by one aggregate score, but it must remain labeled as extension evidence. Its 220-image test partition, source mixture, and retrained checkpoint differ from the primary experiment. Presenting that matrix as the primary SDI-4 result would therefore merge two distinct experimental regimes and overstate what the retained artifacts establish.

A reproducible primary confusion analysis should be regenerated directly from frozen predictions using the exact class order Healthy, BG, WSSV, and WSSV_BG. Both raw counts and row-normalized percentages should be retained, ideally together with confidence scores, so that the actual transition burden and threshold-sensitive mistakes can be inspected without changing the prediction set after the fact. Until those records are available, the report uses the verified aggregate clean-test metrics for the primary conclusion and treats the Section 8 matrix only as supporting evidence. This conservative treatment keeps the later error discussion explicit about which failure patterns are verified and which remain hypotheses for future prediction-level auditing.

4.4 Model Complexity and Efficiency

ASL-LDAM is training-only and adds no inference-time operators. SimAM is parameter-free but not computation-free because it computes feature statistics, a sigmoid map, and element-wise products. The DCFR texture path and channel gate contain trainable convolutions. A matched end-to-end profile of the final CE and final SimAM-DCFR checkpoints under one timing script was not retained, so an exact proposed-overhead delta is not reported.

The latency and FPS values in Table 6.3 are retained auxiliary GPU screening measurements for backbone checkpoints under batch-one inference. The final training and reproducibility narrative uses the accepted T4x2 / dual NVIDIA T4 protocol where applicable, but no unsupported T4x2 inference timing is fabricated here. The Android observation in Section 14 measures a different pipeline that includes device execution and application overhead; neither timing result should be transferred across hardware or tasks.

5. Classification Component Analysis

The retained component records were not all produced under a proven one-factor design, and the final selected checkpoint was distinct from the historical component-run series. Table 6.5 therefore uses the term “recorded component configurations” rather than “strict ablation.” The obsolete fullcombination component value is excluded according to the final evidence policy.

Table 6.5: Verified classification component configurations

Configuration	ASL-LDAM	SimAM	DCFR	Macro-F1 (%)	Interpretation boundary
CE reference	No	No	No	89.02	Controlled reference objective
ASL-LDAM	Yes	No	No	89.91	Loss-side configuration
ASL-LDAM + SimAM	Yes	Yes	No	89.73	Loss plus parameter-free attention
ASL-LDAM + DCFR	Yes	No	Yes	86.76	Loss plus trainable texture recalibration
Final ASL-LDAM + SimAM-DCFR	Yes	Yes	Yes	91.01	Separately selected final checkpoint; not a matched component-run row

Assessment. The component rows are comparative records rather than a strict one-factor ablation; the final checkpoint has the highest recorded Macro-F1, but differing provenance prevents attributing the gain uniquely to one component.

ASL-LDAM alone was 0.89 points above CE in its retained run, while ASL-LDAM + SimAM was 0.71 points above CE. The ASL-LDAM + DCFR configuration was 2.26 points below CE, showing that the texture branch was not independently reliable in that recorded setting. The final selected checkpoint exceeded CE by 1.99 points, but the different provenance prevents attributing that gain uniquely to one component. The evidence supports the final integrated configuration under its recorded protocol; it does not establish an additive contribution model.

6. Classification Error Analysis

Three evidence constraints shape the error analysis. First, the primary prediction file was not retained, so exact error transitions are unavailable. Second, the background-removal process can suppress irrelevant context but can also remove weak edge or texture information. Third, BG, WSSV, and WSSV_BG share visible patterns, and a single image may contain small lesions, pose variation, occlusion, or illumination effects that make the class boundary ambiguous.

The SDI-4 + EXT-3-Original classwise report in Section 8 shows high WSSV precision but lower

WSSV recall, together with high WSSV_BG recall and lower precision. That pattern is consistent with some single-disease images being absorbed by the co-infection class, or with mixed visual evidence being over-predicted. It is a dataset-specific observation rather than proof that the same transition dominated the original 173-image test set.

Qualitative XAI panels indicate that the classifier often responds to shrimp-body regions rather than the entire background. However, some methods produce broader activation than others, and a highlighted region can reflect correlation, image contrast, or model sensitivity rather than a disease mechanism. Future prediction-level failure analysis should combine exact transitions, confidence, original and background-removed images, and multiple explanation methods once the required artifacts are available.

7. Classification Robustness under Controlled Synthetic Corruptions

The clean test images were transformed by five controlled corruption families while labels were preserved. These experiments measure sensitivity to selected synthetic degradations; they are not field validation and do not cover compression, color cast, turbidity, sensor differences, or every farm background.

Table 6.6: Classification robustness under selected synthetic corruptions

Corruption	CE Macro-F1	Proposed Macro-F1	Δ (pp)	CE Kappa $\times 100$	Proposed Kappa $\times 100$
Impulse noise	48.50	55.82	+7.32	41.96	48.44
Gaussian noise	74.79	78.50	+3.71	67.75	71.82
Contrast reduction	82.07	88.10	+6.03	76.90	84.68
Defocus blur	83.00	88.66	+5.66	78.02	85.33
Low light	83.15	90.22	+7.07	77.59	87.11

Assessment. The proposed classifier is higher on all five retained aggregate corruption conditions, but impulse noise remains difficult; a positive delta does not mean the corruption has been solved.

The proposed classifier was higher for all five retained aggregate conditions. The largest displayed Macro-F1 differences occurred under impulse noise (+7.32 points), low light (+7.07), and contrast reduction (+6.03). Impulse noise remained difficult for both models, with the proposed score still far below clean performance. A positive delta therefore does not imply that the corrupted condition was solved.

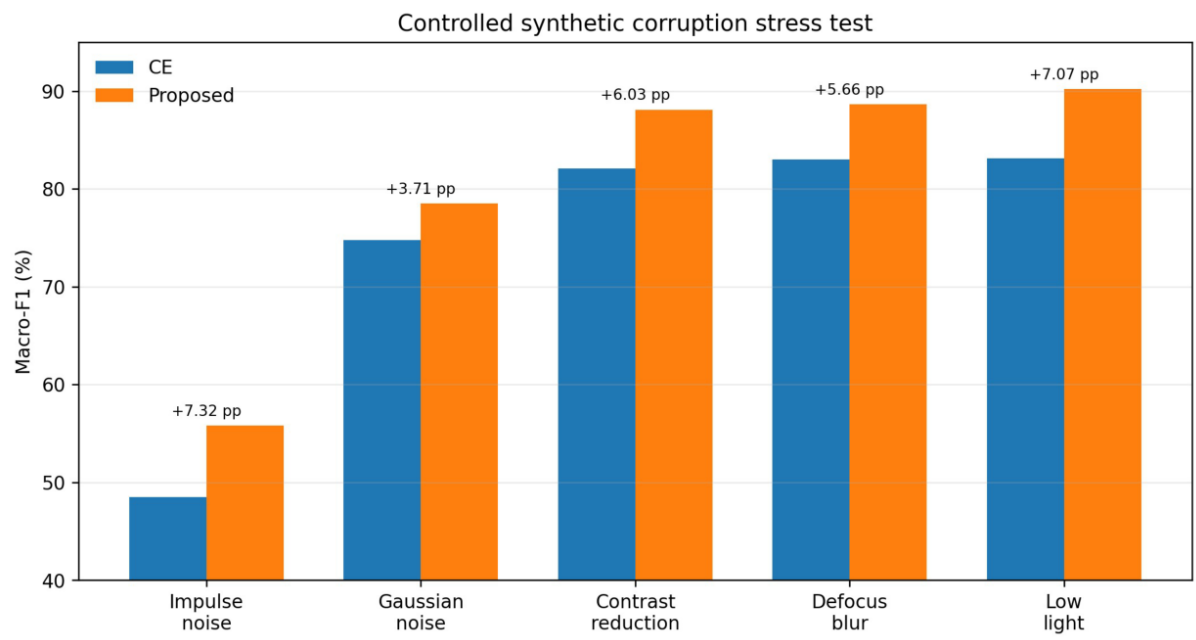


Fig. 6.3. Baseline and proposed Macro-F1 under the five retained classification corruption families.

Impulse Noise — SDI-4 fixed seed-42 test samples
Visualization-strength figures only

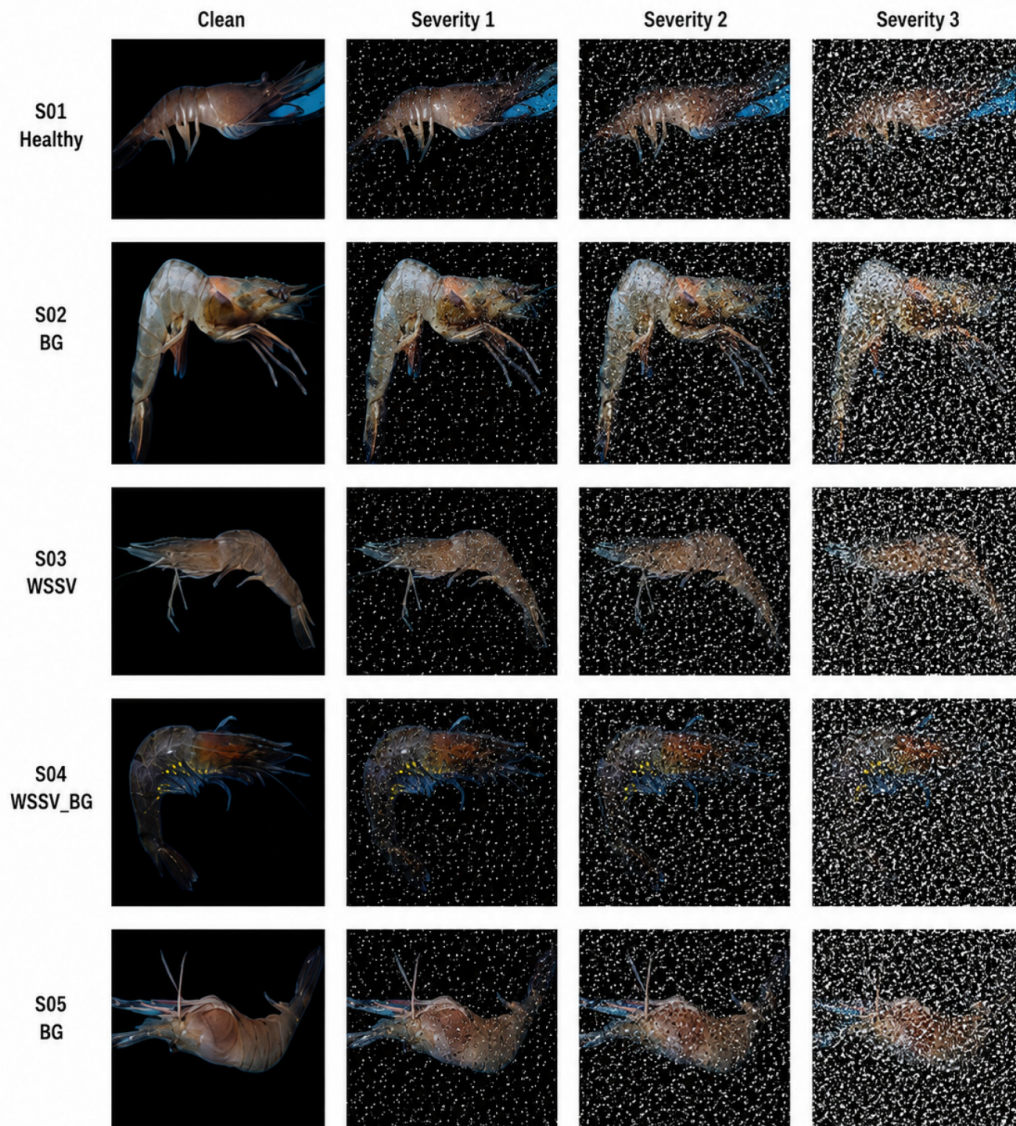


Fig. 6.4. Visualization-strength impulse-noise examples on five fixed seed-42 SDI-4 test samples. Columns show the clean image and three increasingly visible illustration levels. These settings support qualitative inspection and do not replace the retained quantitative corruption protocol.

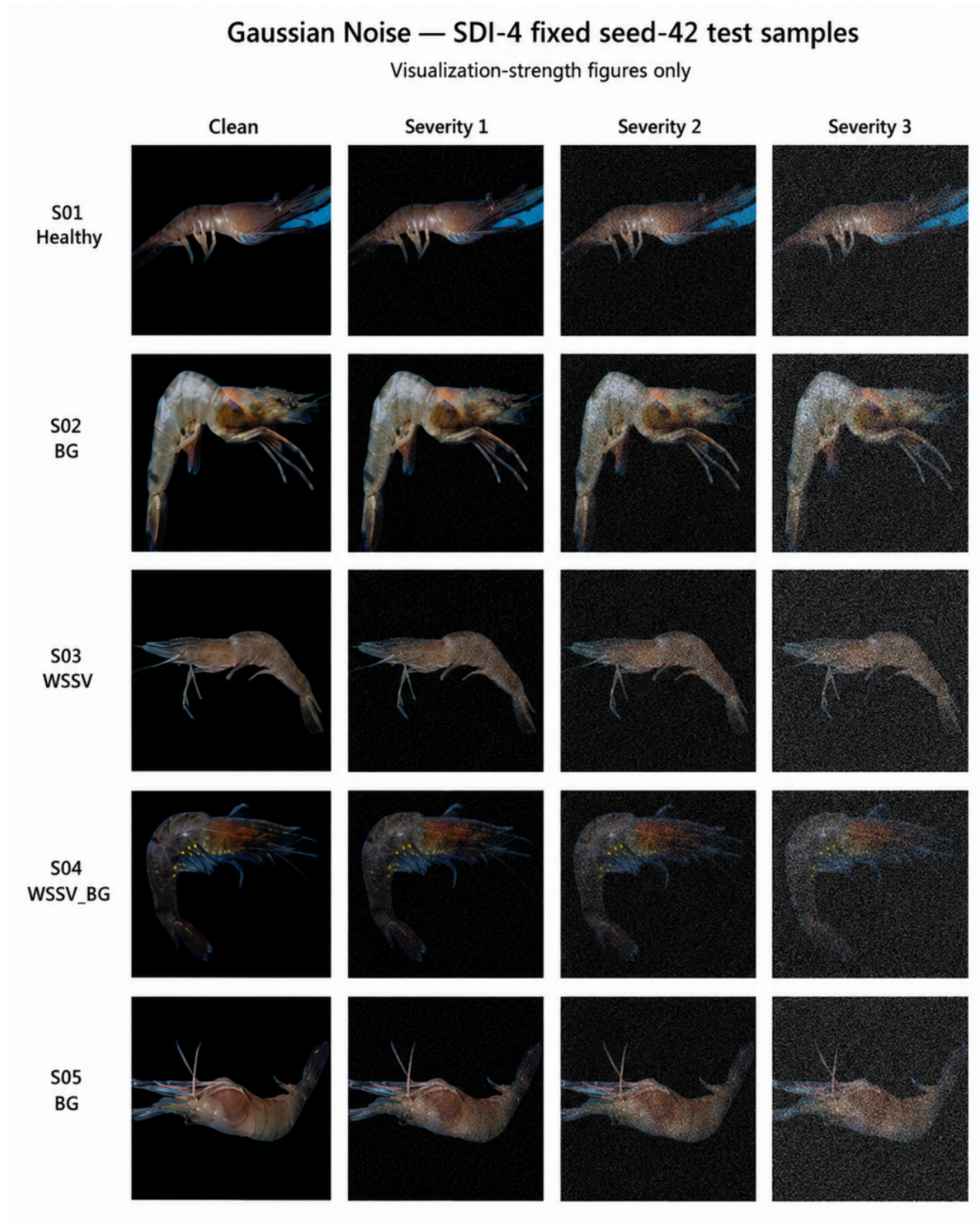


Fig. 6.5. Visualization-strength Gaussian-noise examples on five fixed seed-42 SDI-4 test samples. Columns show the clean image and three increasingly visible illustration levels. These settings support qualitative inspection and do not replace the retained quantitative corruption protocol.

Contrast Reduction — SDI-4 fixed seed-42 test samples
Visualization-strength figures only



Fig. 6.6. Visualization-strength contrast-reduction examples on five fixed seed-42 SDI-4 test samples. Columns show the clean image and three increasingly visible illustration levels. These settings support qualitative inspection and do not replace the retained quantitative corruption protocol.

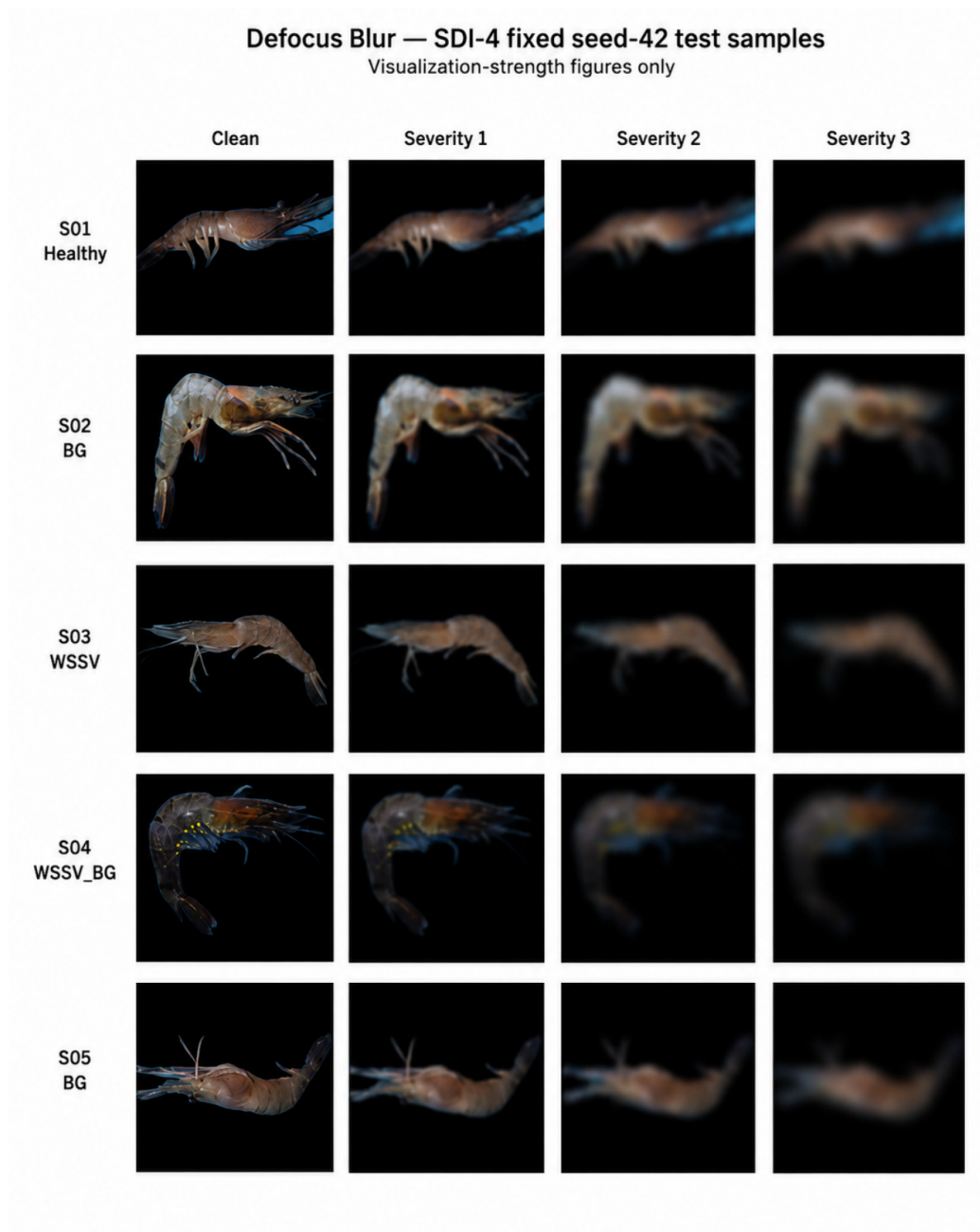


Fig. 6.7. Visualization-strength defocus-blur examples on five fixed seed-42 SDI-4 test samples. Columns show the clean image and three increasingly visible illustration levels. These settings support qualitative inspection and do not replace the retained quantitative corruption protocol.

Low Light — SDI-4 fixed seed-42 test samples
Visualization-strength figures only

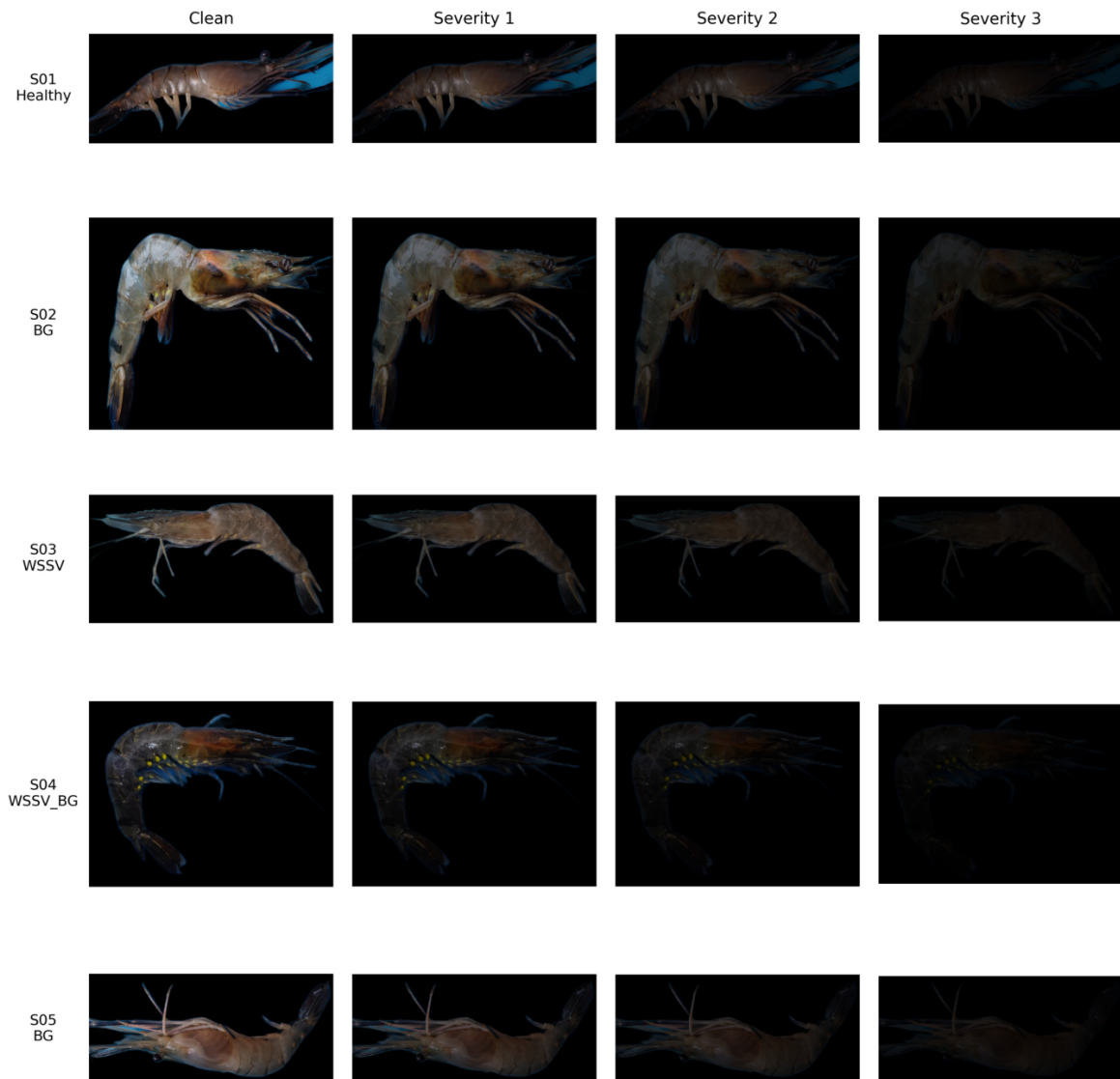


Fig. 6.8. Visualization-strength low-light examples on five fixed seed-42 SDI-4 test samples. Columns show the clean image and three increasingly visible illustration levels. These settings support qualitative inspection and do not replace the retained quantitative corruption protocol.

8. Additional-Source and Partition-Wise Integrated Classification Evaluation

Table 6.7: Classification results on EXT-3-Original and SDI-4 + EXT-3-Original

Dataset regime	Method	Accuracy (%)	Macro-F1 (%)	Test images
EXT-3-Original	YOLO26m-cls+CE	70.21	72.30	47
EXT-3-Original	YOLO26m-cls+ ASL-LDAM+ SimAM-DCFR	91.49	91.29	47
SDI-4+EXT-3-Original	YOLO26m-cls+CE	87.73	86.51	220
SDI-4+EXT-3-Original	YOLO26m-cls+ ASL-LDAM+ SimAM-DCFR	83.18	82.18	220

Assessment. The proposed configuration improves EXT-3-Original but underperforms CE on SDI-4 + EXT-3-Original, showing that the outcome is source-regime dependent rather than uniformly superior.

The opposite outcomes in the two additional-source regimes require a source-aware interpretation rather than the shorthand explanation that one dataset is simply “easier.” EXT-3-Original is a three-class retraining problem without WSSV_BG, whereas the integrated regime restores the four-class task and changes the source mixture, class priors, acquisition backgrounds, and the proportion of each class contributed by each source. WSSV_BG remains supplied only by SDI-4. ASL-LDAM uses class-dependent margins and the final classifier also changes feature recalibration; both mechanisms can interact with a changed empirical class distribution after retraining. However, no matched classwise causal experiment was retained that would isolate one of these factors as the unique cause of the 4.33-point integrated-regime deficit.

The YOLO26m-cls versus YOLO26x-cls train/validation diagnostic in Figure 6.1A provides an additional capacity-related boundary. A substantially larger classifier did not improve validation performance under the same fixed protocol, so the integrated-regime drop should not be attributed to insufficient parameter count alone. The evidence instead supports a more conservative conclusion: performance depends on the joint dataset regime, class composition, optimization objective, and source characteristics, and the proposed configuration is not uniformly superior across retrained source combinations.

On EXT-3-Original, the proposed configuration exceeded CE by 18.99 percentage points in Macro-F1. On SDI-4 + EXT-3-Original, CE exceeded the proposed configuration by 4.33 points. This reversal is central to the interpretation: the integrated method improved the primary SDI-4 clean result and the separately trained three-class external experiment, but it did not dominate the partition-wise integrated four-class regime.

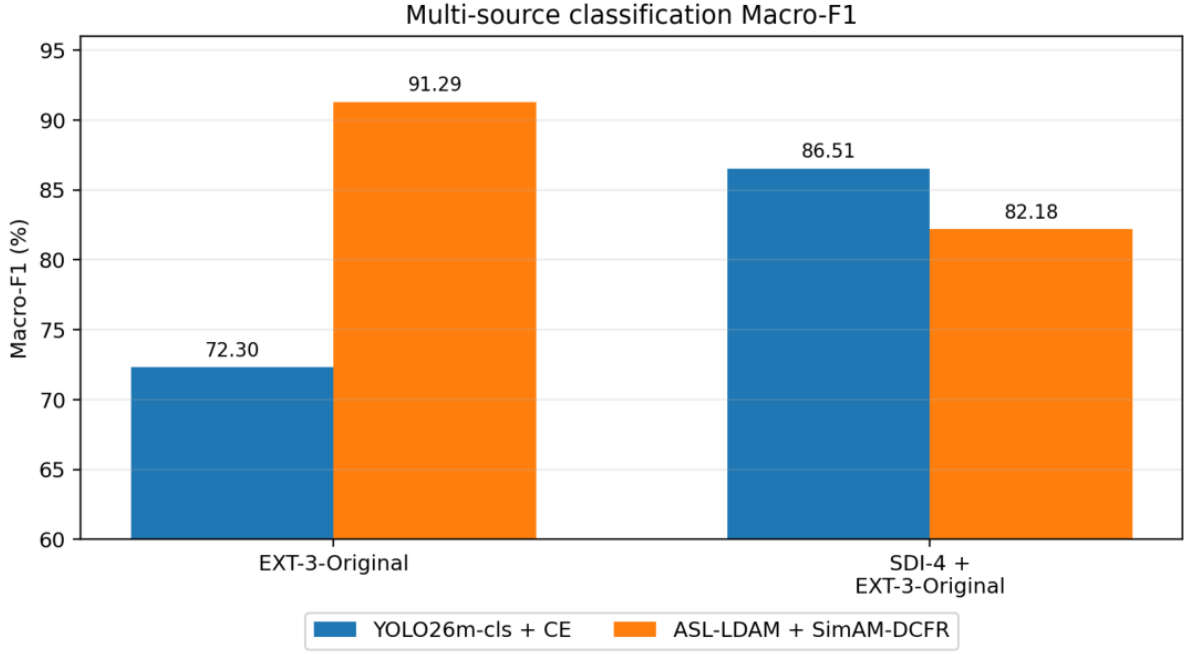


Fig. 6.9. Macro-F1 of CE and the proposed configuration on EXT-3-Original and SDI-4 + EXT-3-Original.

The 47-image EXT-3-Original test partition is sensitive to a small number of prediction changes. The 220-image integrated test partition is larger but changes source composition and class priors. Neither experiment is direct external inference by the original SDI-4 checkpoint.

Table 6.8: Classwise performance of the proposed classifier on SDI-4 + EXT-3-Original

Class	Precision (%)	Recall (%)	F1-score (%)	Support
Healthy	88.00	91.67	89.80	72
BG	80.85	84.44	82.61	45
WSSV	90.91	71.43	80.00	70
WSSV_BG	67.44	87.88	76.32	33
Macro average	81.80	83.85	82.18	220

Assessment. WSSV shows high precision with lower recall, whereas WSSV_BG shows higher recall with lower precision; the integrated-regime aggregate therefore contains a meaningful class-specific precision-recall trade-off.

The classwise report shows that WSSV achieved high precision but lower recall, while WSSV_BG achieved comparatively high recall with lower precision. This pattern is consistent with confusion between disease-specific and co-infection evidence. The table belongs only to the SDI-4 + EXT-3-Original experiment and does not replace the missing classwise ledger for the primary 173-image SDI-4 test partition.

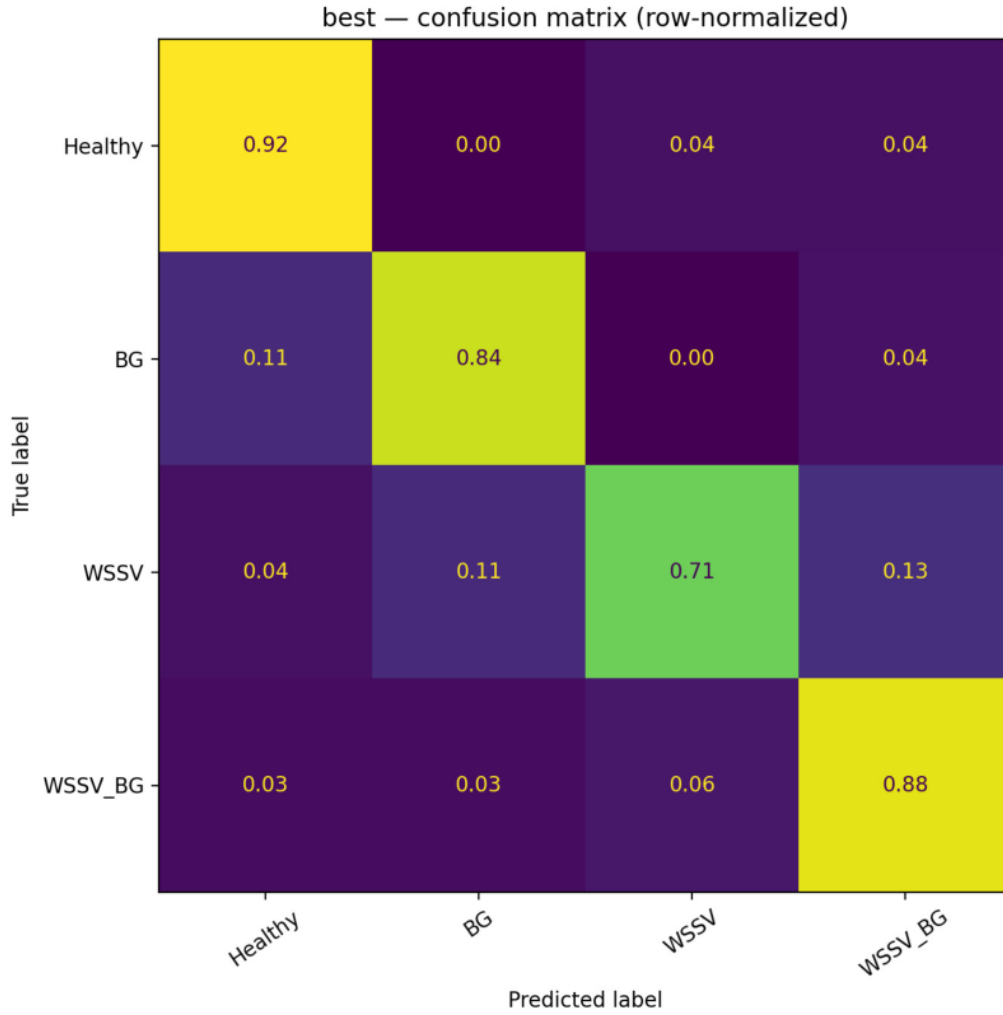


Fig. 6.10. Row-normalized confusion matrix for the proposed classifier trained and evaluated on SDI-4 + EXT-3-Original.

The matrix must not be presented as the primary SDI-4 confusion matrix. It is retained as supporting evidence for the partition-wise integrated regime and demonstrates that aggregate accuracy can conceal class-specific precision-recall trade-offs.

9. Explainability Analysis

Grad-CAM++ [1], HiResCAM [2], and EigenCAM [3] were generated for correctly classified SDI-4 + EXT-3-Original samples. The methods answer related but different questions: gradient-weighted class activation, spatially detailed activation-gradient products, and class-independent principal activation structure. Figure 6.11 places the methods on common samples so that agreement and disagreement can be inspected directly.

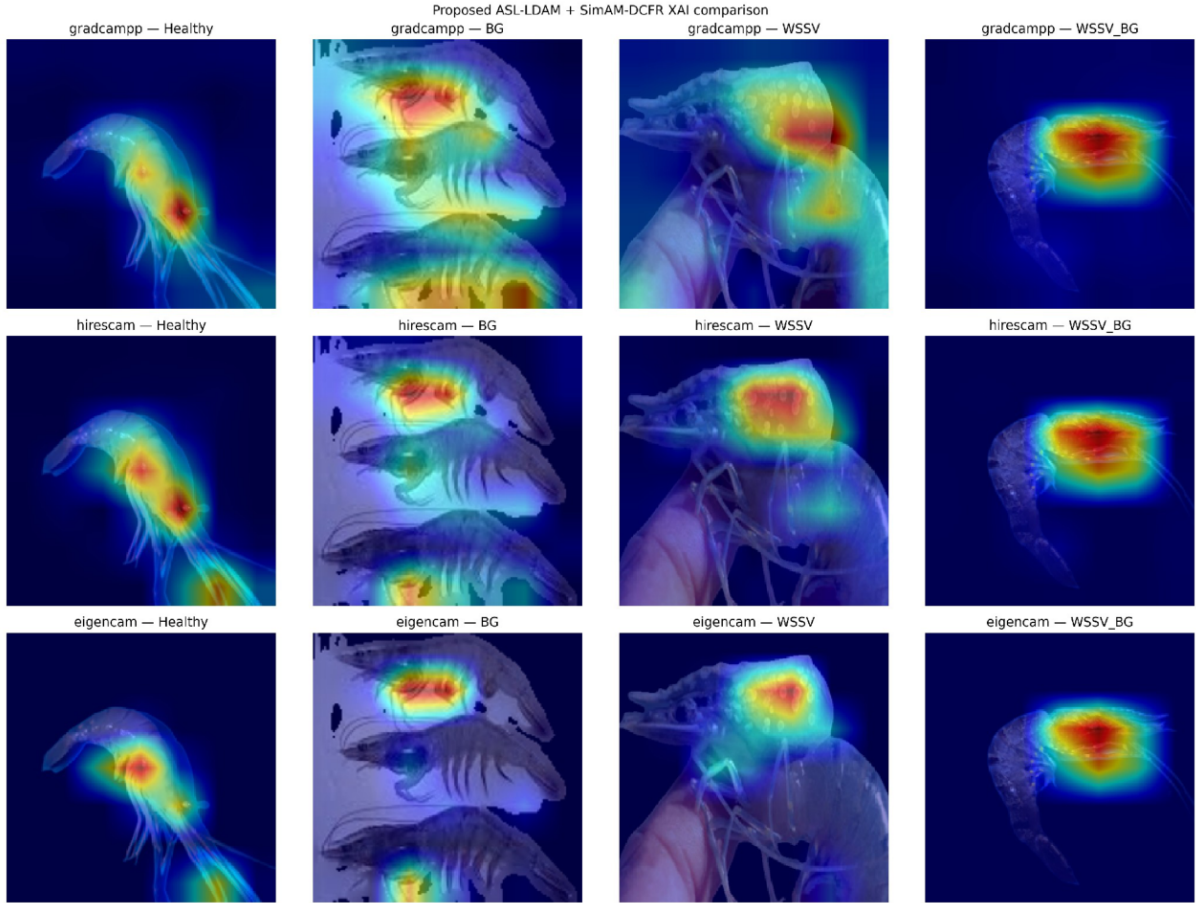


Fig. 6.11. Cross-method qualitative comparison of Grad-CAM++, HiResCAM, and EigenCAM on retained classification samples.

Agreement over shrimp-body regions is useful as a sanity check against exclusive background reliance. Disagreement in spatial extent is expected because the methods aggregate features differently. Broad heatmaps should not be interpreted as segmentation masks, and concentrated activation does not establish causal reliance on a clinically meaningful symptom. The XAI evidence is qualitative, was generated on selected samples, and does not estimate localization accuracy.

10. Instance-Segmentation Baselines and Leakage Analysis

10.1 Segmentation Baseline Comparison

Six Ultralytics segmentation candidates were evaluated under the grouped split. Table 6.9 combines full-test and labeled-only localization with the Healthy false-positive rate and the project-defined HScore. HScore is an internal decision criterion, not an external benchmark.

Table 6.9: Segmentation baseline comparison under the grouped split

Model	Params (M)	Full mAP50	Labeled mAP50	mAP50-95	Healthy FP (%)	HScore (%)
YOLO11n-seg	2.877	48.2	51.2	16.8	24.4	46.0
YOLO26s-seg	11.506	41.7	46.0	18.2	26.8	39.7
YOLOv8n-seg	3.410	39.5	43.2	13.9	34.1	35.3
YOLOv8s-seg	11.821	39.1	44.4	15.9	34.1	36.4
YOLO26n-seg	3.126	43.0	46.3	16.0	19.5	39.3
YOLO11s-seg	10.113	29.7	36.1	12.4	51.2	25.0

Assessment. YOLO11n-seg is retained as the attention reference because it offers the strongest overall balance of localization, compactness, HScore, and Healthy-negative behavior within this baseline set, not because it leads every metric.

Table 6.9B: Additional instance-segmentation comparators on the grouped leakage-safe split

Model	Params (M)	GFLOPs	Full mAP50	Labeled mAP50	Labeled mAP50-95	Recall	Healthy FP (%)
RTMDet-Ins-Tiny	5.615	11.8	44.90	50.90	16.40	64.71 ^a	95.12
RTMDet-Ins-S	10.153	21.5	48.30	54.90	18.70	64.71 ^a	100.00
YOLO26n-seg	3.054	10.3	35.43	36.88	13.00	36.00 ^b	9.76
YOLO26s-seg	11.435	37.3	40.92	44.10	16.70	48.40 ^b	24.39

Assessment. RTMDet-Ins-S provides strong localization but records a 100% Healthy false-positive rate and a much larger parameter count; the comparator therefore exposes a localization-versus-negative-robustness trade-off.

^a RTMDet recall was computed at IoU 0.50 and confidence 0.25. ^b YOLO mask recall is the Ultralytics-exported recall. These recall values are reported for context but are not treated as numerically identical because the evaluation implementations differ. MMEvaluator time for RTMDet is not pure model-inference latency and is therefore excluded from the speed comparison.

RTMDet-Ins-S produced the strongest localization result among the four additional comparators, reaching 48.30% full-test mAP50, 54.90% labeled mAP50, and 18.70% labeled mAP50-95. This gain was accompanied by a 100.00% Healthy false-positive rate in the retained Healthy-negative diagnostic, meaning that every Healthy negative image produced at least one retained disease prediction under that

evaluation. RTMDet-Ins-Tiny showed the same trade-off direction, with 95.12% Healthy FP. In contrast, the original YOLO11n-seg reference in Table 6.9 achieved 48.2% full mAP50 and 51.2% labeled mAP50 with a substantially lower 24.4% Healthy FP rate and only 2.877M parameters. YOLO11n-seg was therefore retained as the attention reference because localization quality, model compactness, and Healthy-negative behavior were more balanced; the selection does not imply dominance on every individual metric.

Because the additional comparator set uses evaluation implementations with non-identical recall definitions, Table 6.9B is interpreted as a separate comparator study rather than a direct extension of the Ultralytics rank order in Table 6.9. The architecture decision is consequently multi-objective: localization performance is evaluated jointly with parameter efficiency and Healthy-negative robustness. RTMDet-Ins-S optimization and validation-based checkpoint selection are examined in Section 10.1.1, and the final architecture trade-off is analyzed in Section 10.1.2.

10.1.1 RTMDet-Ins-S Training and Validation Evidence

RTMDet-Ins-S was trained and evaluated on the same specimen-grouped leakage-safe partition used for the additional comparator study: 331 training groups (905 images, 810 instances), 40 validation groups (115 images, 102 instances), and 45 test groups (129 images, 119 instances). The test partition contained 88 disease-labeled images and 41 Healthy empty-label negatives. The model used 640 x 640 inputs, 100 configured epochs, physical batch size 4, and gradient accumulation of 4, corresponding to an effective batch size of 16. The two predicted instance classes were BG and WSSV.

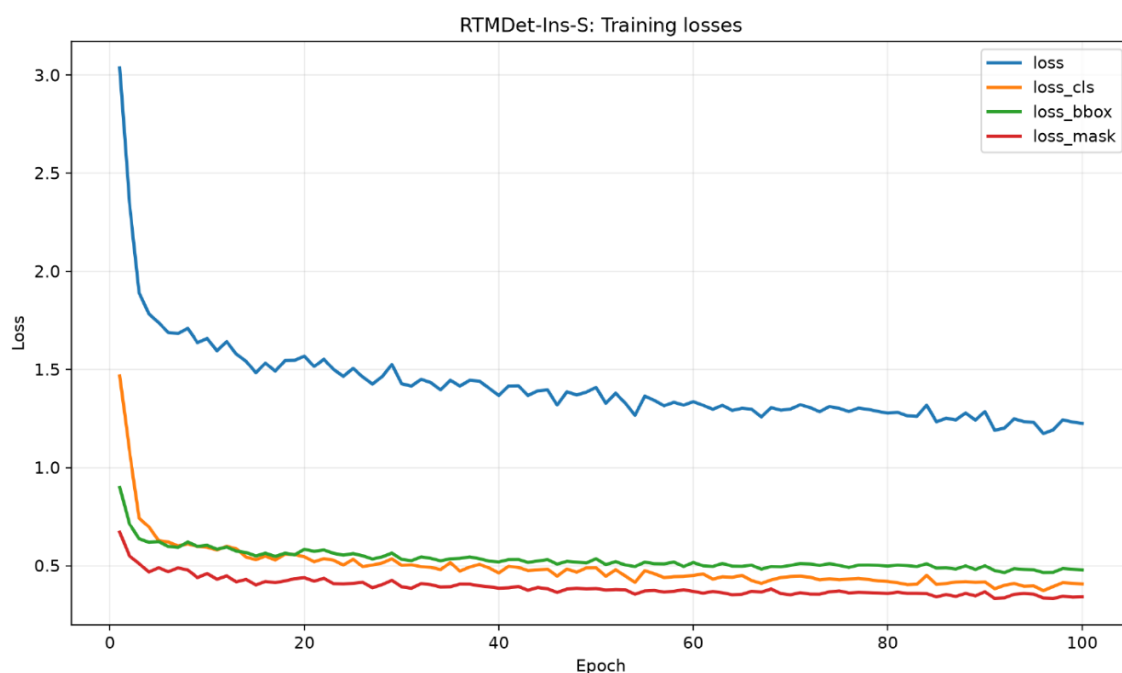


Fig. 6.11A. RTMDet-Ins-S training-loss trajectories over 100 epochs. The total objective and its classification, bounding-box, and mask components are reported from the retained MMDetection log.

The training curves show rapid optimization during the initial epochs followed by a slower decline. The retained log decreases from approximately 3.08 total loss at the beginning of training to approximately 1.23 near epoch 100. Classification loss falls most sharply during the early phase, while bounding-box and mask losses decline more gradually. This behavior is consistent with convergence of the training objective, but it is not by itself evidence of better held-out segmentation quality. The historical MMDetection run used for Table 6.9B did not record validation loss; checkpoint selection therefore relied on validation COCO mask mAP50 rather than on a train-versus-validation loss minimum.

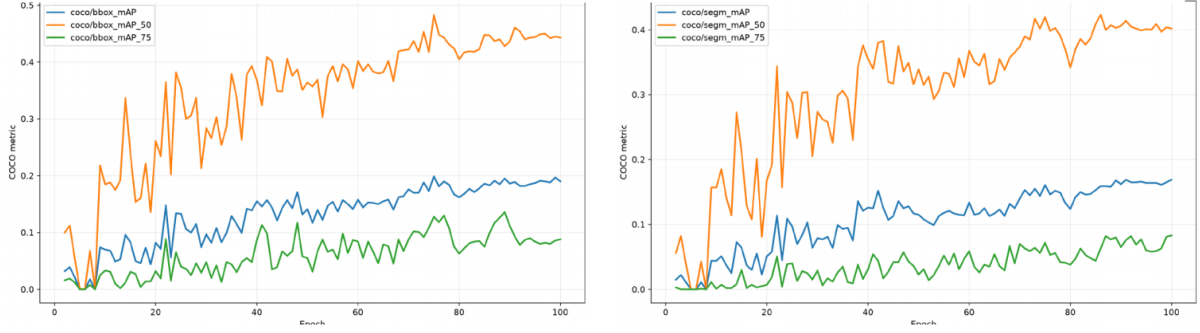


Fig. 6.11B. RTMDet-Ins-S validation COCO trajectories. Left: bounding-box mAP, mAP50, and mAP75. Right: mask mAP, mAP50, and mAP75. These curves are validation AP metrics, not validation-loss curves.

Validation performance was substantially noisier than the training objective during the early epochs and then stabilized. The retained validation mask mAP50 reached its maximum of 42.30% at epoch 86. At that checkpoint, mask mAP50-95 was 15.90% and mask mAP75 was 6.30%; the corresponding bounding-box metrics were 44.70% mAP50, 18.30% mAP50-95, and 9.90% mAP75. By epoch 100, validation mask mAP50 had decreased to 40.20%. Consequently, `best_coco_segmmAP_50_epoch_86.pth`, rather than the final epoch, was used to obtain the RTMDet-Ins-S test results reported in Table 6.9B. This distinction prevents the endpoint test row from being interpreted as an arbitrary epoch-100 result.

10.1.2 Selection of YOLO11n-seg over RTMDet-Ins-S

RTMDet-Ins-S is a strong localization comparator, but model selection in this study is multi-objective rather than mAP-only. Relative to the original YOLO11n-seg grouped baseline, RTMDet-Ins-S improves full-test mAP50 only marginally (48.30% versus 48.2%) while increasing labeled mAP50 from 51.2% to 54.90% and labeled mAP50-95 from 16.8% to 18.70%. The principal failure is Healthy-negative robustness: RTMDet-Ins-S produced at least one retained disease prediction on all 41 Healthy test images, corresponding to a 100% Healthy false-positive rate, whereas YOLO11n-seg recorded 24.4%. The comparator is also substantially larger, with 10.153M parameters compared with 2.877M for YOLO11n-seg. Thus, the higher disease-image localization score did not compensate for the 75.6-percentage-point deterioration in Healthy false-positive rate and the markedly larger model footprint.

The later CA → SimAM refinement further strengthens this engineering choice: the selected YOLO11n-seg + CA → SimAM model uses 2.855M parameters and approximately 9.7 GFLOPs while reaching 55.07% labeled mAP50 and 19.64% labeled mAP50-95 (Table 6.21), both slightly above the RTMDet-Ins-S labeled metrics. Because framework implementations and recall definitions are not perfectly identical, this comparison is not presented as a universal architecture ranking. Within the retained CVio evaluation, however, YOLO11n-seg provides the more appropriate foundation for a lightweight mobile-oriented system because it combines competitive localization, substantially safer Healthy-negative behavior, and much lower complexity.

YOLO11n-seg had the highest full and labeled mAP50 and HScore in this baseline table. YOLO26s-seg had the highest mAP50-95, while YOLO26n-seg had the lowest Healthy false-positive rate. Thus, the baseline screen did not yield one model that dominated every criterion. YOLO11n-seg was retained as the attention reference because its overall localization-efficiency trade-off was strongest within this recorded set.

10.2 Split-Policy Leakage Comparison

Table 6.10: Leakage comparison across segmentation split policies

Split policy	Train-test overlap (%)	Full mAP50	Labeled mAP50	mAP50-95	Healthy FP (%)
Plain random image	98.7 ± 1.2	67.1 ± 1.3	67.7 ± 1.3	29.9 ± 2.4	5.4 ± 2.7
Stratified random image	94.0 ± 6.1	65.1 ± 6.0	65.9 ± 6.2	28.2 ± 1.6	9.8 ± 2.4
Stratified grouped specimen	0.0 ± 0.0	45.1 ± 2.7	47.3 ± 3.5	14.4 ± 2.1	23.6 ± 11.0

Assessment. Image-level random splits produce larger mAP together with very high specimen overlap, whereas the grouped specimen split removes train-test group overlap and is the principal leakage-validity reference.

Plain random splitting exceeded grouped splitting by 20.4 points in labeled-only mAP50 while retaining almost complete train-test specimen overlap. Bold in the overlap column marks the validity objective rather than a performance ranking. The lower grouped result is not evidence that grouping damages the model; instead, it indicates that repeated-view leakage made the random-split problem easier. The grouped protocol is the principal evaluation because it better approximates performance on previously unseen filename-derived specimens.

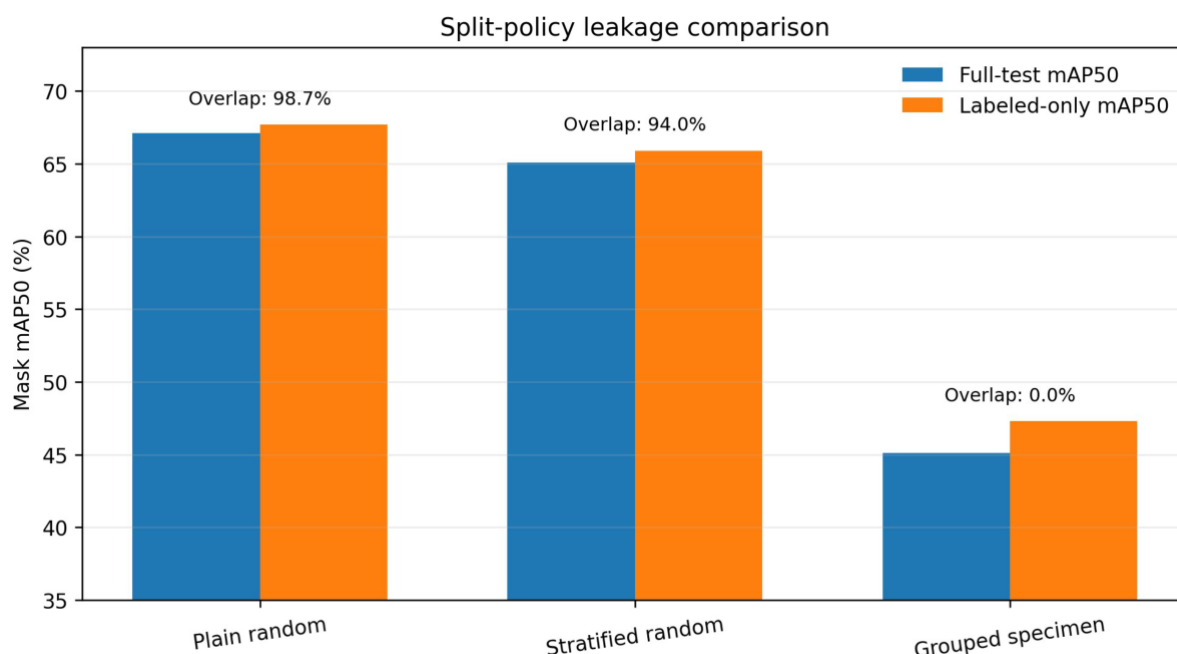


Fig. 6.12. Full-test and labeled-only mask mAP50 under image-level and grouped-specimen split policies.

The grouped split also produced a higher and more variable Healthy false-positive rate, exposing a failure mode hidden by the leaky splits. This supports reporting negative-image diagnostics together with mAP rather than selecting a split solely because it yields larger scores.

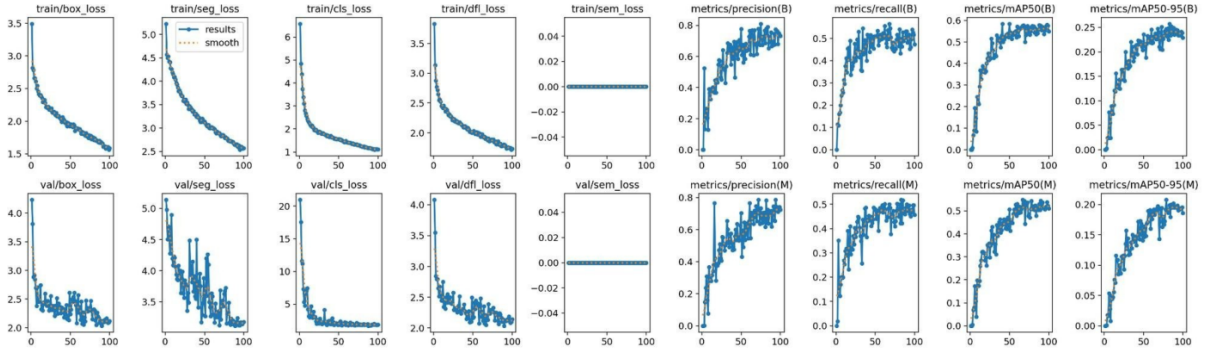


Fig. 6.13. Representative training and validation curves retained for convergence inspection.

11. Attention-Based Instance-Segmentation Results

The attention runs were not fully identical. They differed in augmentation, epoch budget, early stopping, software versions, module placement, and whether an auxiliary loss was introduced. Table 6.11 preserves these differences rather than presenting the study as a perfect one-factor ablation.

Table 6.11: Recorded attention configuration protocols

Configuration	Placement/objective	Aug. policy	Configured/executed epochs	Init. source	Ultralytics/PyTorch
YOLO11n-seg reference	No added attention	Clean-light	100/60, early stop	yolo11n-seg.pt	8.4.61/ 2.10.0
CA to SimAM	P3, P4, P5	Strong	100/100	594/594 tensors	8.4.67/ 2.11.0
LKA to SimAM	P3, P4, P5	Clean-light	100/75, early stop	579/579 tensors	8.4.61/ 2.10.0
DPCA	P3, P4, P5	Strong	200/200	yolo11n-seg.pt	8.4.102/ 2.10.0
CoTEGate + BoundaryLite	P4 + auxiliary loss	Strong	200/184, early stop	yolo11n-seg.pt	8.4.95/ 2.10.0

Assessment. Because augmentation, training budget, early stopping, initialization, software versions, placement, and objectives differ across rows, this table supports configuration-level comparison rather than a strict causal ablation.

Table 6.12: Attention localization and efficiency summary on the original grouped dataset

Model	Full mAP50	Labeled mAP50	Labeled mAP50-95	Recall	Parameters	Inf. ms/image	FPS
YOLO11n-seg baseline	44.09	46.64	16.90	41.0	2,842,998	6.0	166.7
CA → SimAM	49.71	55.07	19.64	49.7	2,854,718	9.3	107.5
LKA → SimAM	50.03	52.49	18.12	54.4	2,963,510	8.6	116.3
DPCA	48.80	52.25	15.80	51.3	2,848,377	8.8	113.6
CoTEGate + BoundaryLite	49.21	51.74	16.60	49.7	2,736,562	8.2	122.0

Assessment. CA to SimAM leads labeled mAP50 and labeled mAP50-95 with only 11,720 additional parameters over the reference; LKA to SimAM leads full-test mAP50 and recall, while the unmodified baseline remains fastest overall.

The efficiency columns were added to answer the Council's request for parameter-level comparison. CA → SimAM achieved the highest labeled mAP50 (55.07%) and labeled mAP50-95 (19.64%), while increasing the reference parameter count by only 11,720 parameters. LKA → SimAM led full-test mAP50 and recall but had the largest parameter count in this attention set. CoTEGate + BoundaryLite used the fewest parameters and was the fastest attention variant in the retained notebook timing, whereas the unmodified baseline remained fastest overall. The recorded millisecond/FPS values are notebook GPU measurements and are not Android benchmarks. Because augmentation, training budget, initialization, software versions, placement, and auxiliary losses differ across rows, the table supports configuration-level comparison rather than a strict one-factor causal ablation.

Table 6.12B: Placement ablation for CA → SimAM on YOLO11n-seg

Placement	Full mAP50	Labeled mAP50	Labeled mAP50-95	Recall	Parameters	Inf. ms/image	FPS
P3/P4/P5	49.71	55.07	19.64	49.7	2,854,718	9.3	107.5
Neck concat	40.90	43.30	14.20	46.5	2,818,718	10.3	97.1
Head P3	40.90	43.40	14.60	41.5	2,691,582	9.9	101.0
Head P4	47.70	50.50	16.80	47.4	2,693,246	9.6	104.2
After C2PSA	43.80	45.80	16.90	45.6	2,696,574	10.3	97.1
Backbone P3	38.50	40.90	13.60	43.6	2,691,582	10.6	94.3
Neck P3/P4	44.30	47.30	15.90	43.6	2,721,250	10.4	96.2

Assessment. P3/P4/P5 is the selected placement because it leads all retained localization and recall columns in this controlled placement study; Head P4 is the strongest single-placement alternative.

The P3/P4/P5 placement led all four retained localization/recall columns in this placement study. Among single placements, Head P4 was the strongest localization choice, but it remained below the

multi-level P3/P4/P5 configuration. The result supports the selected multi-scale placement within this controlled placement experiment; it does not convert the broader attention-family comparison into a strict ablation.

Table 6.13: Healthy-negative and mask-count diagnostics on the original grouped dataset

Model	Healthy FP images(%)	FP masks	Disease misses (%)	Full MAE	Labeled MAE	HScore
Reference	15 (36.59)	17	11 (12.50)	0.354167	0.354167	0.3934
CA to SimAM	14 (34.15)	16	12 (13.64)	0.333333	0.306818	0.4808
LKA to SimAM	15 (36.59)	18	7 (7.95)	0.475452	0.492424	0.4518
DPCA	11 (26.83)	12	12 (13.64)	0.313953	0.323864	0.4590
CoTEGate + BoundaryLite	15(36.59)	17	10(11.36)	0.395349	0.386364	0.4540

Assessment. CA to SimAM has the highest HScore, DPCA the lowest Healthy false-positive rate, and LKA to SimAM the lowest disease missed-image rate; no configuration dominates every diagnostic axis.

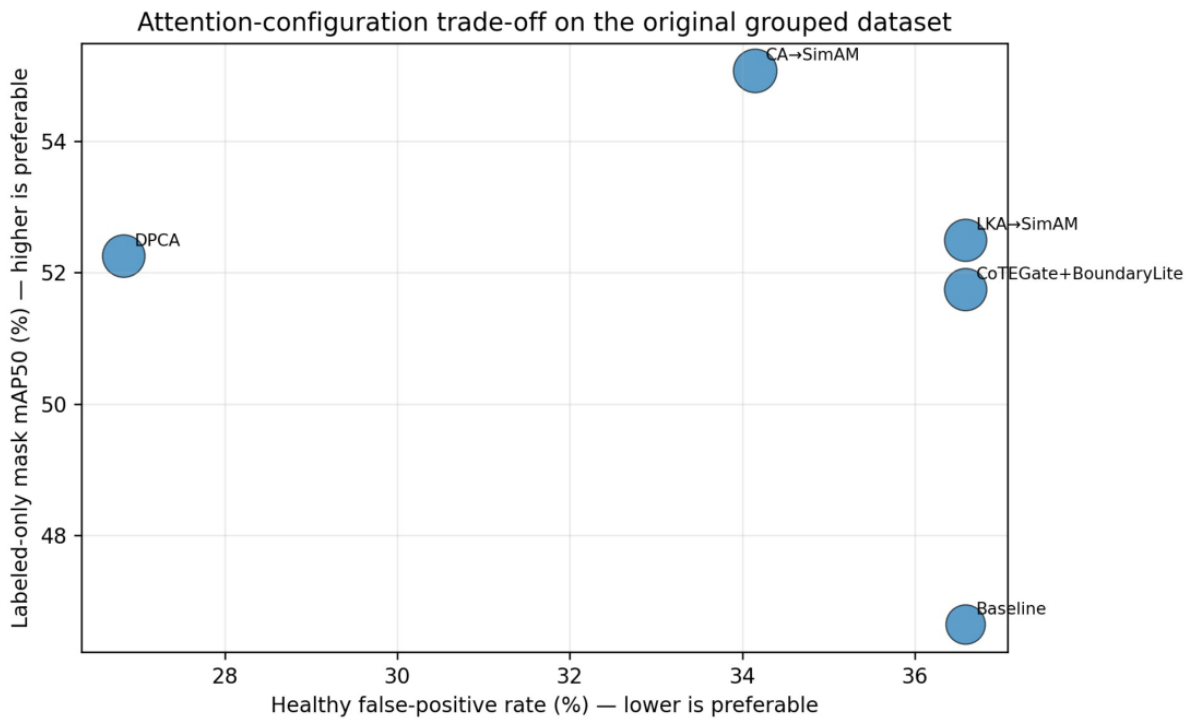


Fig. 6.14. Trade-off between labeled-only mAP50, Healthy false-positive rate, and HScore for the original grouped attention runs.

CA to SimAM had the highest HScore, DPCA the lowest Healthy false-positive rate, and LKA to SimAM the lowest disease missed-image rate. No method dominated all diagnostic axes. CoTEGate + BoundaryLite changed both architecture and training objective, so its row is not an attention-only estimate.

Table 6.14: Attention results on the combined segmentation dataset

Model	Full mAP50	Labeled mAP50	Labeled mAP50-95	Healthy FP (%)	HScore
CA to SimAM	62.99	65.94	25.72	39.0	0.5913
LKA to SimAM	56.79	58.50	25.02	39.0	0.5110
DPCA	50.46	53.14	19.15	48.8	0.4471
CoTEGate + BoundaryLite	43.99	48.00	17.33	36.6	0.3966

Assessment. CA to SimAM leads the recorded localization metrics and HScore on the combined inventory, but the changed dataset and mixed grouped-stratified allocation prevent interpreting this as a direct attention-only comparison with the original grouped study.

The combined dataset contained 1,452 images and used a mixed grouped-stratified allocation because 303 additional diseased images did not follow the original filename convention. CA-to-

SimAM led the recorded localization metrics and HScore in this extension, but the changed dataset and split prevent a direct attention-only comparison with the original grouped table.

12. Fourier-Domain Instance-Segmentation Results

12.1 Fourier Candidate Screening

The strict no-augmentation, hook-off screen isolated image-space frequency preprocessing. Each model received the intended transformed evaluation input. Table 6.15 reports the complete retained candidate table.

Table 6.15: Fourier candidate screening under the strict control

Configuration	Labeled mAP50	mAP50-95	Healthy FP (%)	Disease miss (%)	Healthy-aware score (%)
No Fourier control	23.54	6.06	39.02	22.73	14.08
High-pass $\sigma = 50$, $\alpha = 0.05$	30.88	7.92	34.15	25.00	20.90
W1 high-pass $\sigma = 50$, $\alpha = 0.10$	34.51	8.85	63.41	19.32	21.08
High-frequency damping	32.11	10.68	39.02	52.27	17.40
Band-pass 12/60, $\alpha = 0.20$	29.52	8.39	26.83	43.18	17.21
Low-frequency flat-tening	27.87	8.52	14.63	46.59	16.66
Homomorphic filtering	26.12	9.16	41.46	23.86	15.06
Gaussian low-pass $\sigma = 25$	25.08	8.56	31.71	37.50	15.37

Assessment. W1 has the highest labeled mAP50 and Healthy-aware score in the isolated screen, but its 63.41% Healthy false-positive rate is severe; it is therefore retained as an interaction-study reference rather than a universal preprocessing winner.

W1 had the highest labeled mAP50 and Healthy-aware score in this isolated screen, improving mAP50 by 10.97 points. However, its Healthy false-positive rate increased to 63.41%, which is operationally severe. High-frequency damping achieved the highest mAP50-95 but missed more than half of the diseased images. The screen therefore selected W1 as a high-pass reference for interaction studies, not as a universally superior preprocessing method.

12.2 W1 High-Pass Reference

W1 is the retained image-space high-pass reference with a Gaussian low-pass scale of $\sigma = 50$ and residual enhancement $\alpha = 0.10$. The transform reconstructs a smooth low-frequency component and adds a scaled residual back to the original image before clipping. Because the operation changes pixel intensities without geometric warping, the image dimensions and polygon coordinates remain aligned. This made W1 suitable for controlled interaction experiments in which any change in segmentation behavior should arise from appearance-frequency modification rather than from moving the annotated disease regions.

Its selection as the reference was deliberately narrower than choosing it as the best preprocessing method. In the strict no-augmentation, hook-off screen, W1 reached 34.51% labeled mAP50 and the highest retained Healthy-aware score of 21.08%, but the Healthy false-positive rate rose to 63.41%. The same configuration missed 19.32% of disease-labeled images. These values show a genuine sensitivity-versus-specificity trade-off: emphasizing high-frequency residuals can make disease boundaries or fine texture more detectable, while also amplifying benign edges, shell texture, sensor noise, or other structures that can resemble disease evidence on Healthy negatives.

12.2 W1 High-Pass Reference (continued)

The fact that W1 helped most strongly under the strict control is also part of its interpretation. When conventional augmentation and the hidden Albumentations hook were disabled, the high-pass transform supplied an appearance perturbation that was otherwise absent. Once spatial and photometric augmentation were introduced, the incremental value of W1 became regime-dependent rather than consistently positive. This is why Section 12.3 studies augmentation order and hook state explicitly instead of treating the +10.97-point strict-control gain as an additive improvement that should transfer to every training policy.

W1 is therefore retained as an interaction-study reference and as a mechanism-oriented probe, not as the final deployment preprocessing path. The later paired spectrum analysis examines whether W1 measurably shifts low-, mid-, and high-frequency energy under matched augmentation realizations, while the segmentation metrics determine whether those spectral changes are useful or harmful. Even there, the evidence remains empirical and bounded: clipping and image quantization introduce nonlinearities, and a frequency-energy shift does not by itself establish a causal disease feature. The operational conclusion is simply that frequency enhancement can change the localization/negative-control balance and must be evaluated together with augmentation and Healthy-image behavior.

12.3 Fourier--Augmentation Order Study

Table 6.16: Interaction between W1 and augmentation regimes

Training regime	No Fourier mAP50	W1 mAP50	Δ (pp)	Healthy FP: off to W1	Healthy-aware: off to W1
No augmentation, hook off	23.54	34.51	+10.97	39.02 to 63.41	14.08 to 21.08
No augmentation, hook on	30.61	29.94	-0.67	36.59 to 41.46	19.73 to 17.53
Clean-light, hook off	47.34	40.82	-6.52	39.02 to 36.59	39.48 to 33.49
Clean-light, hook on	51.20	53.18	+1.98	24.39 to 24.39	45.96 to 47.02
Strong, hook off	59.51	57.71	-1.80	34.15 to 31.71	53.26 to 50.15

Assessment. W1 is beneficial in some augmentation and hook regimes and harmful in others; the strong-augmentation result is highest without W1, so the Fourier effect is interaction-dependent rather than consistently additive.

The effect of W1 depended on conventional augmentation. It improved the strict control and the clean-light hook-on condition, but reduced mAP50 under the clean-light hook-off and strong policies. Strong augmentation without W1 produced the highest labeled mAP50 in this table. This interaction rejects a simple claim that Fourier preprocessing consistently adds to spatial and photometric augmentation.

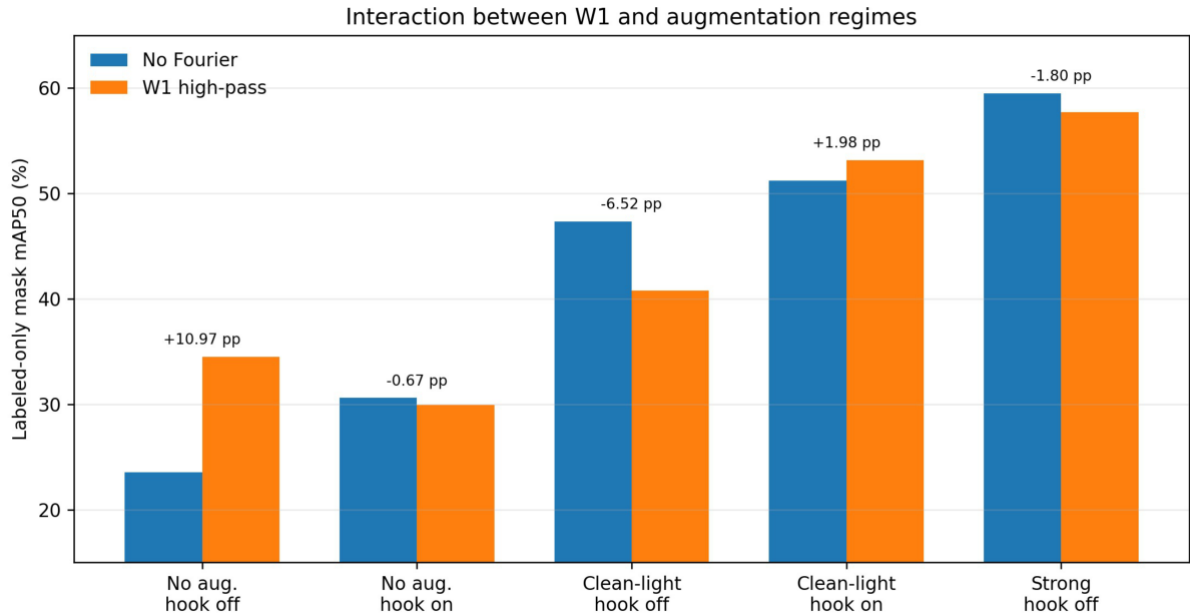


Fig. 6.15. Labeled-only mask mAP50 with and without W1 across five augmentation and hook regimes.

The better order depended on augmentation type. W1 before spatial augmentation was favorable for FlipLR, rotation, and erasing, whereas W1 after augmentation was favorable for FlipUD. Translate and scale remained worse than their augmentation-only references.

Table 6.17: Offline augmentation-Fourier order study

N=1 augmentation	Augmentation-only mAP50	Higher Fourier order	Fourier mAP50	Δ (pp)
Translate 0.08	37.82	Augmentation $\rightarrow$ W1	35.00	-2.82
Scale 0.10	36.91	Augmentation $\rightarrow$ W1	31.85	-5.05
FlipLR 0.25	32.75	W1 $\rightarrow$ augmentation	36.51	+3.77
Rotation 3 degrees	31.08	W1 $\rightarrow$ augmentation	35.95	+4.87
Erasing 0.15	19.81	W1 $\rightarrow$ augmentation	29.07	+9.26
FlipUD 0.30	33.45	Augmentation $\rightarrow$ W1	34.51	+1.05

Assessment. The favorable Fourier order depends on the augmentation transform; the large relative gain for erasing starts from a weak baseline and does not make that configuration the strongest overall.

A large relative gain from a weak baseline, such as erasing 0.15, did not make that configuration the strongest overall model.

Direct Spectrum Analysis of the W1-Augmentation Interaction

The Council specifically asked why W1 improved the no-augmentation control but became slightly harmful under the strong-augmentation policy. The revised analysis therefore uses direct spatial- and frequency-domain evidence rather than inferring the mechanism from mAP alone. The retained paired study used 24 test images (six each from Healthy, BG, WSSV, and WSSV_BG) and 12 matched stochastic augmentation draws per image, yielding 288 paired observations. W1 was evaluated both alone and in combination with the recorded strong augmentation pipeline.

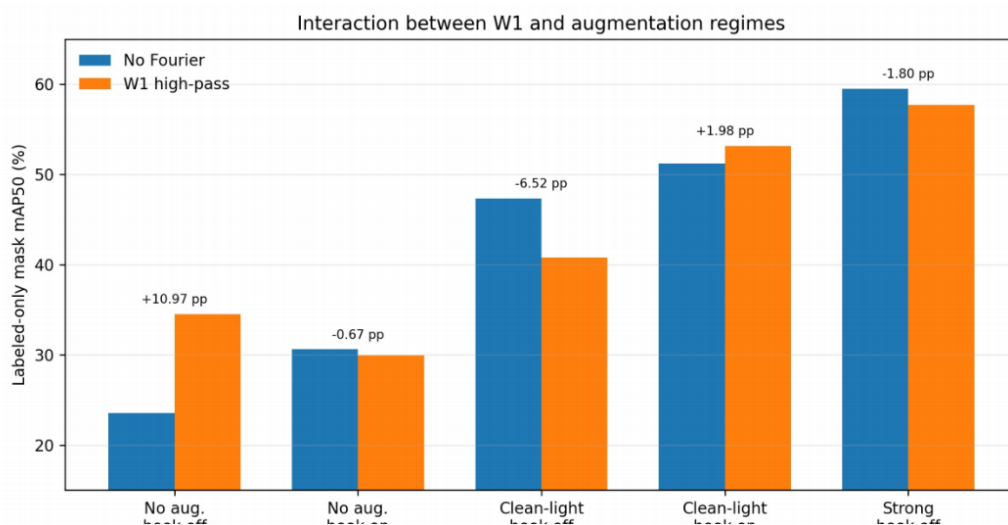


Fig. 6.15A. Labeled-only mask mAP50 with and without W1 across the retained augmentation/hook regimes. W1 improved the strict no-augmentation, hook-off control from 23.54% to 34.51% (+10.97 pp), but reduced the strong-augmentation, hook-off result from 59.51% to 57.71% (-1.80 pp).

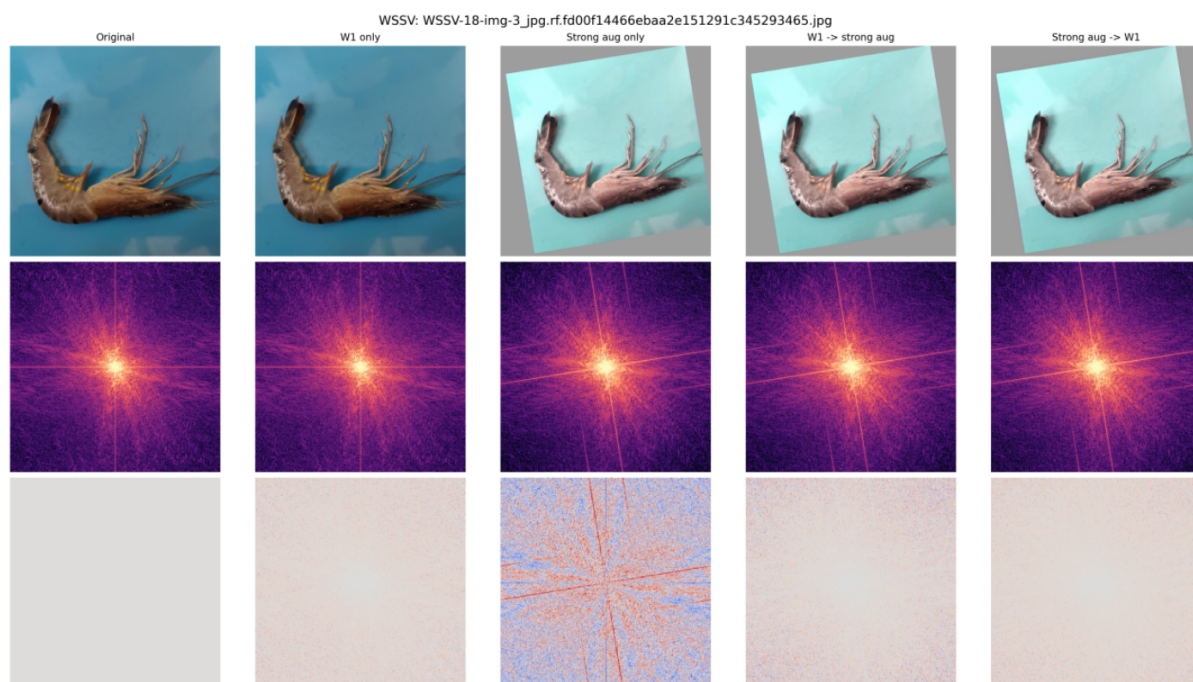


Fig. 6.15B. Representative spatial and Fourier-domain comparison of Original, W1 only, Strong augmentation only, W1 followed by strong augmentation, and strong augmentation followed by W1. The log-power spectra and signed spectral differences show that the augmentation pipeline already redistributes substantial frequency content, while W1 adds a smaller residual emphasis when combined with the stronger transform sequence.

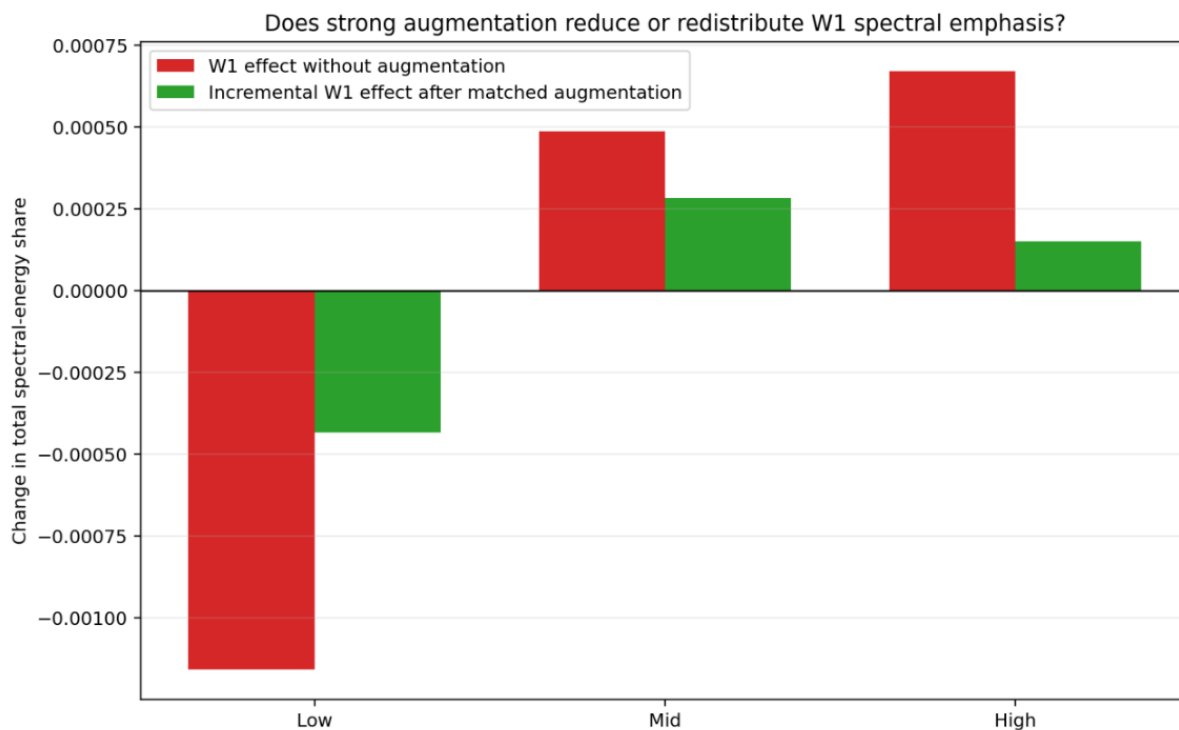


Fig. 6.15C. Mean change in low-, mid-, and high-frequency energy share caused by W1 alone versus the incremental W1 effect after matched strong augmentation. The high-band increment after augmentation is much smaller than the W1-only increment.

Across the 288 paired observations, W1 alone increased the high-band energy share by an average of 0.000670, whereas the incremental increase attributable to W1 after strong augmentation was 0.000151, only 22.5% of the original W1 high-band effect. The retained edge-sensitive summaries showed the same attenuation pattern: the incremental Laplacian-variance effect was 21.1% of the W1-alone effect, mean-gradient effect was 56.1%, and lesion-region gradient effect was approximately 79.0%. These measurements indicate that strong spatial/color augmentation does not completely eliminate W1, but it overlaps with or redistributes much of W1's global high-frequency emphasis. In the recorded pipeline, geometric transforms can resample already enhanced edges and HSV operations alter channel-intensity relationships. This explanation is consistent with the observed spectrum and seed-42 mAP interaction; it is not a multi-seed causal proof.

12.4 Feature-Domain Fourier Refinement

The feature-domain FDDEM direction was implemented separately from W1 input preprocessing. The retained implementation description specifies learnable complex gains in radial bands and residual feature reconstruction, but it does not contain a final, fully audited result table tied to one frozen checkpoint. No numerical FDDEM claim is therefore inserted. This omission preserves the distinction between a described architectural experiment and a verified selected outcome.

Table 6.18: Recorded Fourier and augmentation inference timing

Method	Fourier preprocessing (ms/image)	YOLO prediction (ms/image)	Effective total (ms/image)
W1 high-pass	48.62	38.77	87.38
Erasing0.10	0.00	12.69	12.69
FlipUD 0.30	0.00	12.69	12.69
Scale 0.10+Erasing0.10	0.00	12.82	12.82

Assessment. W1 adds substantial preprocessing cost in this CUDA segmentation timing scope, increasing effective total latency to 87.38 ms/image; these values are not Android runtime and are not interchangeable with classification timing.

The timing protocol used the same 129-image grouped test set, CUDA execution, 640-pixel inputs, ten warm-up images, and 119 timed images. W1 added 48.62 ms of preprocessing and increased effective latency to 87.38 ms per image in that segmentation timing scope. This measurement is not interchangeable with the auxiliary classification screening latency or the Android runtime because the task, model, hardware context, and measured pipeline differ.

13. Segmentation Robustness and Failure Analysis

The segmentation robustness set differed from the classification corruption set. The retained positive-gain table contains color cast, turbidity, JPEG artifacts, overexposure, and Gaussian noise. Because the rows were selected by positive improvement, they cannot establish that the proposed segmentation configuration improved every evaluated corruption.

Table 6.19: Selected attention-based segmentation corruption results

Corruption	Reference la- beled mAP50(%)	Attention-based proposed la- beled mAP50(%)	Improvement (pp)
Color cast	29.81	54.65	+24.84
Turbidity	25.03	44.97	+19.94
JPEG artifact	36.58	56.39	+19.81
Overexposure	35.21	52.46	+17.25
Gaussian noise	18.13	33.44	+15.31

Assessment. The table contains selected positive-gain segmentation corruptions rather than a complete benchmark; color cast has the largest displayed improvement, while Gaussian noise remains comparatively difficult.

Table 6.19 belongs to the attention-based segmentation branch. Color cast produced the largest selected difference, followed by turbidity and JPEG artifacts. Gaussian noise remained difficult even after improvement. The table reports selected positive-gain conditions rather than a complete corruption benchmark; Fourier preprocessing is evaluated separately in Section 12 and is not merged into the attention-based column.

Color cast

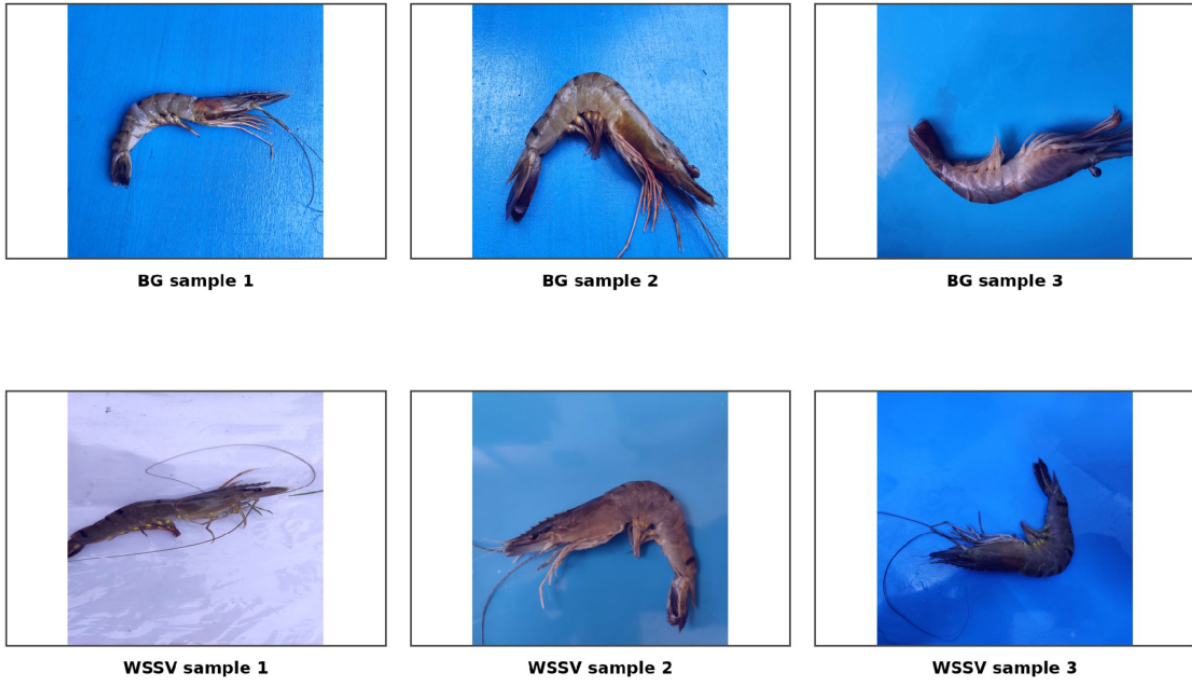


Fig. 6.16. Representative color-cast segmentation examples for BG and WSSV. The original class and annotation semantics are retained; the panel is qualitative and does not quantify mask accuracy.

Turbidity-style degradation

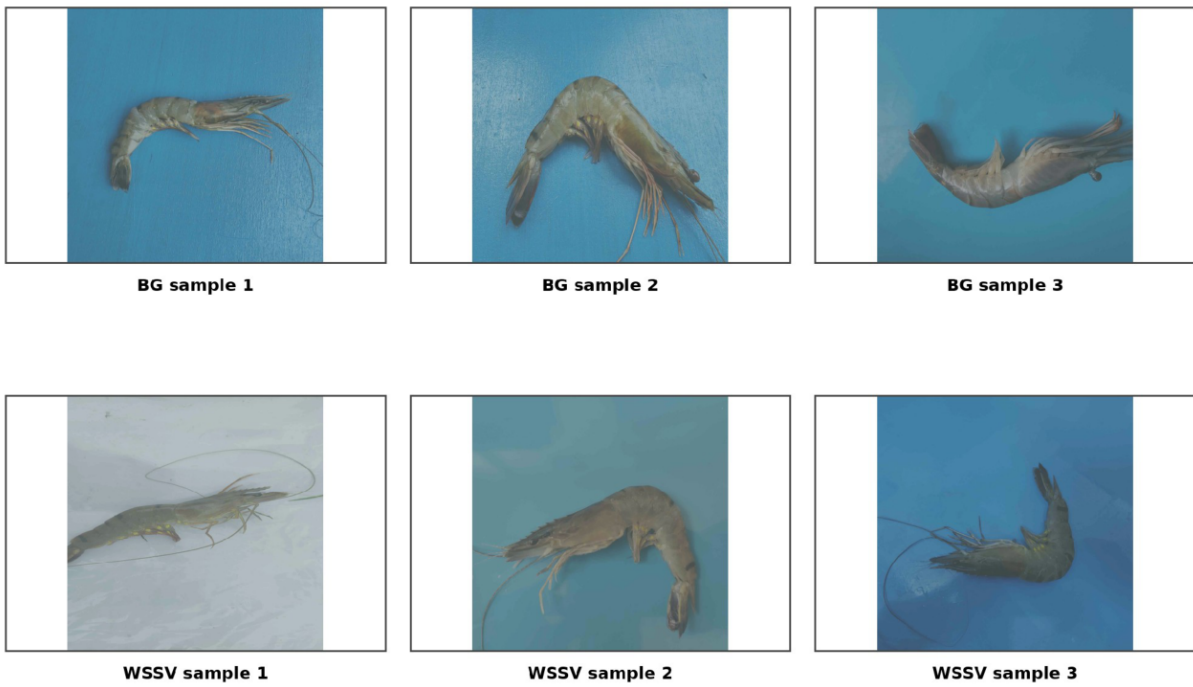


Fig. 6.17. Representative turbidity-style segmentation examples for BG and WSSV. The original class and annotation semantics are retained; the panel is qualitative and does not quantify mask accuracy.

JPEG artifact

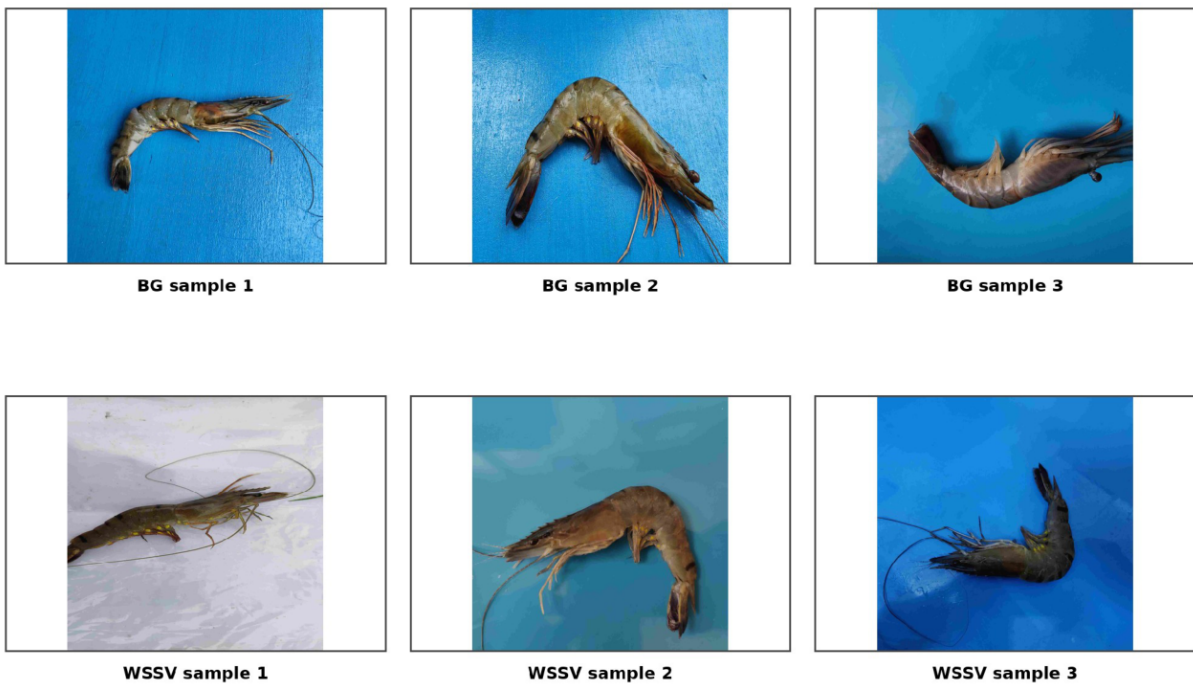


Fig. 6.18. Representative JPEG-artifact segmentation examples for BG and WSSV. The original class and annotation semantics are retained; the panel is qualitative and does not quantify mask accuracy.

Overexposure

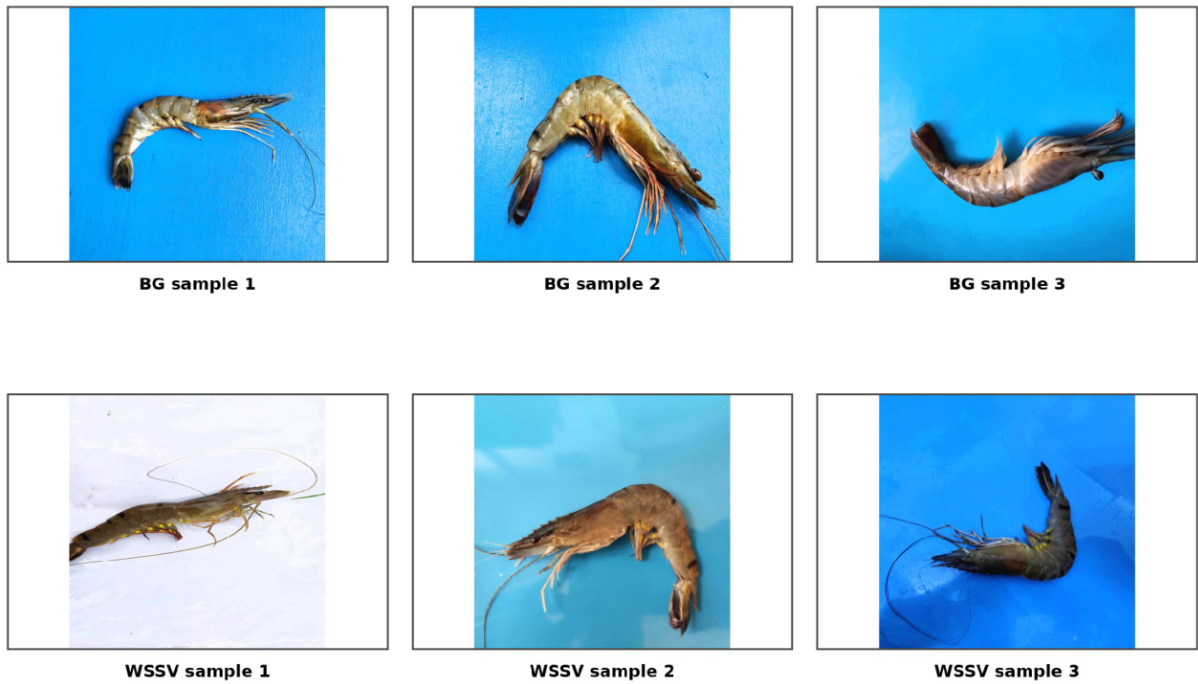


Fig. 6.19. Representative overexposure segmentation examples for BG and WSSV. The original class and annotation semantics are retained; the panel is qualitative and does not quantify mask accuracy.

Gaussian blur

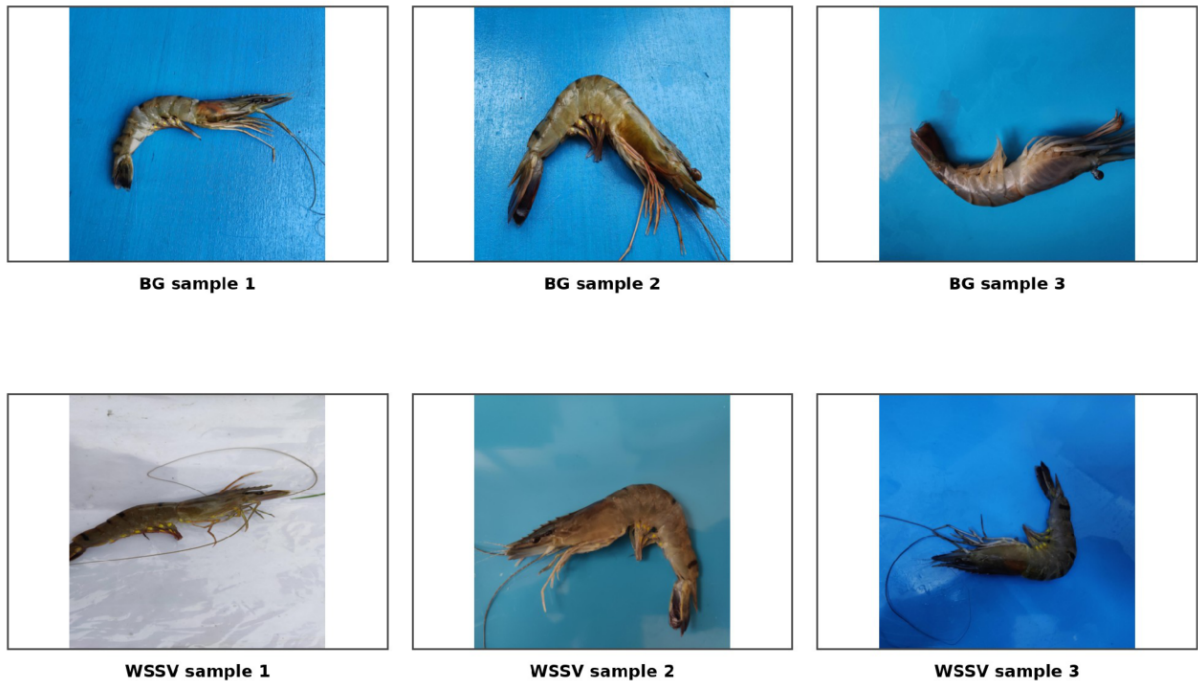


Fig. 6.20. Representative Gaussian-blur segmentation examples for BG and WSSV. This retained source panel represents blur and is distinct from the Gaussian-noise metric row in Table 6.19; it is qualitative and does not quantify mask accuracy.

Figures 6.16–6.19 illustrate the first four metric-matched conditions in Table 6.19. Color cast and turbidity alter global appearance, JPEG artifacts introduce block and ringing structure, and overexposure

removes contrast. The retained fifth image source, shown in Figure 6.20, is explicitly Gaussian blur, whereas the fifth quantitative row is Gaussian noise. Because these are distinct transformations, the blur panel is retained as separate qualitative evidence and is not used to illustrate or validate the Gaussian-noise metric.

Common segmentation failure mechanisms include missed small lesions, fragmented masks, mask spill beyond the visible symptom, Healthy false positives, and ambiguous overlap between BG and WSSV evidence. The Figure 6.21 panels below support qualitative inspection of these behaviors but are not the held-out test aggregate and do not quantify boundary error.

14. Mobile Deployment Observations

The revised mobile evaluation distinguishes model compactness from on-device runtime. The implemented Android workflow is a conditional two-model pipeline. Every image first passes through the FP32 classifier `yolo26m_asl_ldam_simam_dcfp32.tflite`. If the classifier returns Healthy above the application threshold, the workflow stops after classification. If it returns BG, WSSV, or WSSV_BG above threshold, the application additionally invokes the FP16 instance-segmentation artifact `yolo11n_seg_float16.tflite`. Low-confidence outputs also terminate without segmentation. Therefore a single “mobile inference time” must not be interpreted as the cost of one fixed model path.

Table 6.20: Final conditional Android inference contract

Item	Implemented / retained evidence	Interpretation boundary
Stage 1 classifier	YOLO26m-cls + ASL-LDAM + SimAM-DCFR; 10,886,467 parameters; FP32 TFLite; input [1, 224, 224, 3]; output [1, 4]	Runs for every image; ASL-LDAM is training-only
Dispatch rule	Healthy or below-threshold output: no segmentation; BG/WSSV/WSSV_BG above threshold: invoke segmentation	Predicted branch is an application control signal, not verified disease ground truth
Stage 2 segmentation	YOLO11n-seg + CA → SimAM at P3/P4/P5; FP16 TFLite, 5.65 MB	Runs conditionally for disease-triggered outputs
Execution scope	Android CPU path; four threads in the configured application	No iOS/iPhone test; no verified cross-device GPU/NNAPI comparison
Logged output	Predicted class, confidence, branch-dependent runtime, history record	Ground-truth fields are empty in the field runtime logs; no mobile accuracy is reported
Timing boundary	Application-recorded elapsed time for the operational pipeline	Not isolated pure network latency; may include preprocessing, interpreter/postprocessing, and application overhead

Assessment. The deployed Android pipeline is conditional: classification always runs, segmentation runs only for above-threshold disease outputs, and the recorded application time is therefore branch-dependent rather than pure single-model latency.

Table 6.21: Final selected instance-segmentation model summary

Item	Verified information
Final segmentation model	YOLO11n-seg + CA → SimAM at P3/P4/P5
Input	640 × 640 RGB
Parameters	2,854,718 (2.855M)
Complexity	Approximately 9.7 GFLOPs
Artifact size	best.pt approximately 6.1 MB; yolo11n_seg_float16.tflite 5.65 MB
Key localization results	Full-test mask mAP50 49.71%; labeled mask mAP50 55.07%; labeled mask mAP50-95 19.64%; full-test recall 49.70%
Notebook timing	9.3 ms/image; 107.5 FPS on the retained notebook GPU timing
Selection rationale	Highest labeled-test localization in the retained attention set with low model complexity and only a small parameter increase over the YOLO11n-seg reference

Assessment. The selected segmenter combines 2.855M parameters and approximately 9.7 GFLOPs with 55.07% labeled mAP50 and 19.64% labeled mAP50-95; the 9.3 ms/image figure is notebook GPU timing, not phone latency.

The 9.3 ms/image value in Table 6.21 is a notebook GPU measurement and is deliberately separated from the Android results below. A model can be compact in parameters and serialized size without achieving the same speed on every phone.

Table 6.22: Android devices used for cross-device runtime profiling

Member	Device	Android	SoC	CPU cores	RAM as reported	Storage
Tran Huu Nhan	vivo Y100	15	Snapdragon 685, up to 2.8 GHz	8	16 GB reported	128 GB
Le Thi Huynh Nhu	Samsung A32	13	MediaTek Helio G80, up to 2.00 GHz	8	6 + 6 GB	128 GB
Nguyen Van Phong	POCO M7 Pro 5G	16	MediaTek Dimensity 7025 Ultra	8	8 + 4 GB	256 GB

Assessment. The three Android devices provide heterogeneous deployment coverage, but the hardware ledger is descriptive: RAM is preserved as reported and missing GPU identifiers are not inferred.

The RAM values are recorded as reported by the devices/users. Values such as 6+6 GB and 8+4 GB include vendor memory-extension configurations and should not be read as wholly physical RAM. GPU model fields were not retained in the supplied runtime records and are therefore not inferred.

Table 6.23: Observed runtime of the conditional Android pipeline on three devices

Device	All runs	No-seg branch, mean	Seg-triggered runs	Seg-triggered branch, mean	Overall mean
vivo Y100	21	14; 5.016 s mean	7	6.914 s	5.649 s
Sam-sung A32	21	2; 0.499 s mean	19	2.155 s	1.997 s
POCO M7 Pro 5G	21	13; 1.860 s mean	8	2.641 s	2.157 s

Assessment. Observed device means depend on different branch compositions and image workloads; the table supports runtime-feasibility evidence but not a strict same-image hardware ranking.

^a The two Samsung A32 no-seg observations were below-threshold short-circuit cases; no Healthy above-threshold record was present in that log. They are therefore not used for direct Healthy-branch hardware ranking. On the vivo Y100, eight Healthy above-threshold runs averaged 5.024 s; on the POCO M7 Pro 5G, ten Healthy above-threshold runs averaged 1.903 s.

All three Android devices used the same application build, the same FP32 classification artifact, and the same FP16 YOLO11n-seg artifact under the configured four-thread CPU execution path. Each member recorded 21 runs using manually uploaded real-world shrimp photographs. Image order was not standardized because the experiment was designed as a runtime-feasibility assessment rather than a controlled image-by-image hardware benchmark. Device-level means are therefore interpreted together with the observed branch composition and are not used as a strict hardware ranking.

The branch-specific results explain why disease-triggered cases can take longer than Healthy or other no-seg cases. Classification is executed first; an above-threshold disease output then dispatches the instance-segmentation model, whereas a Healthy or below-threshold result terminates after classification. On the vivo Y100, the segmentation-triggered branch averaged approximately 1.90 s longer than the aggregate no-seg branch; on the POCO M7 Pro 5G, the corresponding difference was approximately 0.78 s. The Samsung A32 log contains 19 segmentation-triggered records and only two below-threshold no-seg short-circuit cases, so its 0.499 s no-seg mean is not treated as a stable classification-only hardware estimate.

The three-device study evaluates runtime feasibility, not field diagnostic accuracy. The photographed shrimp did not receive veterinary, laboratory, or other expert disease ground truth; accordingly, the CSV ground-truth and correctness fields remain empty. Predicted class labels are used only to determine which application branch executes and are not treated as verified disease labels. No mobile accuracy, sensitivity, specificity, or disease-prevalence result is therefore calculated. iOS was not tested because no compatible iPhone/iOS device was available, and this remains a deployment limitation rather than an inference from the Android results.

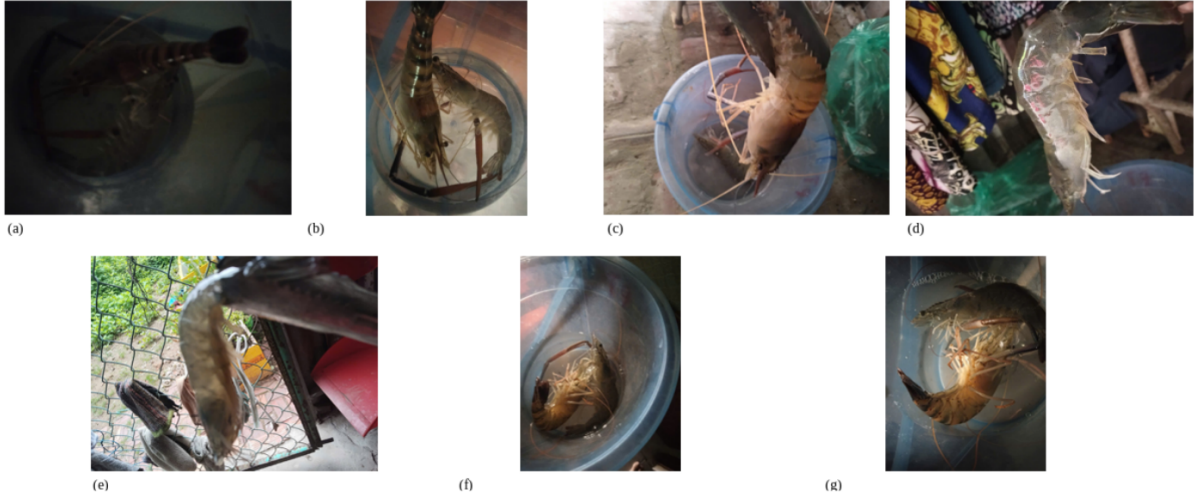


Fig. 6.20A. Representative real-world shrimp photographs from the 21-image operational acquisition set captured using the three team-owned Android phones (vivo Y100, Samsung A32, and POCO M7 Pro 5G). The examples illustrate low illumination, uneven lighting, camera softness/blur, cluttered backgrounds, close-range framing, orientation changes, and varied viewpoints. These photographs support operational runtime testing only; they are not an expert-labeled field-accuracy benchmark.

The field photographs provide a closer operational analogue to the synthetic robustness motivation by exposing the application to naturally occurring capture variation from three Android cameras. They demonstrate that the deployed pipeline was exercised on non-dataset photographs under imperfect acquisition conditions. Because no veterinary or laboratory diagnosis was available, however, the photographs support deployment feasibility and capture-diversity discussion only; they do not establish disease-recognition accuracy or clinical generalization in real-world conditions.

Camera-resolution metadata were not retained sufficiently for a controlled resolution-specific comparison; therefore, the study reports three distinct phone-camera sources without claiming a measured performance effect attributable specifically to camera resolution.

15. Integrated Discussion

The primary classification result demonstrates an observed improvement under the recorded fixed-seed protocol. The 1.99-point Macro-F1 increase matters because the classes were imbalanced and visually overlapping classes require more than aggregate accuracy for interpretation. Kappa moved in the same direction as accuracy and Macro-F1, indicating that the primary improvement was not merely an increase in raw agreement. Nevertheless, the absence of a retained primary classwise report prevents a complete account of which classes contributed most.

The additional-source experiments place an important boundary around the main result. The proposed configuration was much stronger than CE on EXT-3-Original but weaker than CE on SDI-4 + EXT-3-Original. This is evidence that loss design and feature recalibration interact with source composition, class priors, and visual domain. A method that improves one controlled dataset should not be described as uniformly superior across source combinations.

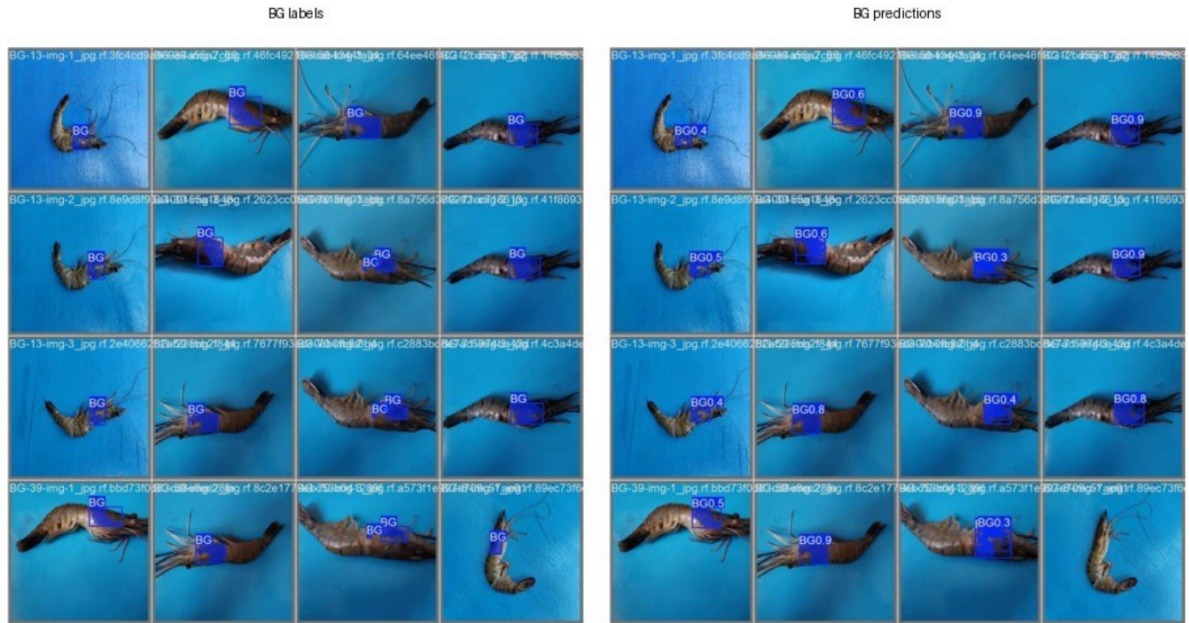


Fig. 6.21(a). BG ground-truth labels and corresponding BG predictions.



Fig. 6.21(b). Healthy empty-label negatives and corresponding disease-mask predictions.



Fig. 6.21(c). WSSV ground-truth labels and corresponding WSSV predictions.

Fig. 6.21. Retained instance-segmentation ground-truth and prediction panels for BG, Healthy, and WSSV evaluation images. BG and WSSV rows compare project-authored disease-region ground-truth masks with predicted masks. Healthy rows are negative controls with no positive disease-mask target; every predicted BG or WSSV region in those rows is therefore a false positive.

The Healthy panel is a negative-control comparison. Healthy images contain no positive BG or WSSV disease-mask target, so every retained predicted disease region in the corresponding prediction panel is a false positive. This qualitative evidence explains why Healthy-negative diagnostics are reported separately from positive-mask average precision.

The corruption results support reduced sensitivity under five selected classification degradations, but the absolute impulse-noise score remained low. The segmentation corruption table was selected by positive gains and used a different set of transformations. These two branches therefore cannot be merged into one universal robustness claim. A unified benchmark with the same severity definitions and reporting policy would be needed for a system-level comparison.

XAI contributes a qualitative sanity check rather than a localization metric. Agreement among Grad-CAM++, HiResCAM, and EigenCAM over shrimp-body regions can reveal obvious background dependence, while disagreement shows that explanation is method-dependent. Neither a heatmap nor a segmentation mask establishes causality or veterinary validity.

The leakage analysis is one of the strongest validity findings. Random image splitting produced much larger mask mAP while nearly identical specimens crossed partitions. The grouped protocol reduced apparent performance but provided a more credible estimate for previously unseen filename-derived specimens. Lower grouped mAP is therefore more informative than higher leaky mAP for the intended validity question.

The attention results show metric-specific trade-offs rather than one dominant configuration. CA to SimAM led labeled localization and HScore, LKA to SimAM led full-test mAP50 and disease miss rate, and DPCA minimized Healthy false positives. Protocol differences prevent assigning these changes solely to attention design. CoTEGate + BoundaryLite is especially non-isolated because it changes both the P4 feature path and the training loss.

The Fourier experiments expose a similar trade-off. W1 substantially improved the strict no-augmentation control but amplified Healthy false positives and imposed a large preprocessing cost. Under strong conventional augmentation, W1 slightly reduced mAP50. The defensible conclusion is that Fourier preprocessing is condition-dependent and must be evaluated with both localization and negative-image guardrails.

The revised deployment evidence now covers three Android devices and the complete classifier-plus-conditional-segmentation control flow. The key distinction is branch-dependent runtime: Healthy and below-threshold outcomes stop after classification, whereas disease-triggered outputs execute the additional FP16 YOLO11n-seg pipeline. This makes a single device-average latency dependent on workload composition. The real-world photographs were captured using the three team cameras under varied acquisition conditions, but because no veterinary/laboratory ground truth was available, they support operational robustness discussion and runtime profiling only, not field accuracy. No iOS device was evaluated.

Relative to the literature reviewed earlier in the thesis, the contribution is not a claim of superiority over incomparable published percentages. The stronger contribution is the combination of a controlled classification reference, explicit imbalance-aware and feature-recalibration design, synthetic corruption testing, multi-source qualification, specimen-grouped segmentation, Healthy-negative diagnostics, attention/Fourier trade-off analysis, and a traceable mobile interface. Each claim remains bounded by its dataset, split, checkpoint, and artifact provenance.

16. Threats to Validity and Limitations

The classification evidence uses one fixed seed, so run-to-run variance and statistical significance are unknown. Test-partition Macro-F1 was inspected during candidate screening, weakening model-family ranking independence. Complete primary classwise predictions and a primary confusion matrix were not retained, which limits exact error attribution. Dataset coverage remains constrained by public sources, class imbalance, acquisition bias, and background removal. The mapped EXT-3-Original and SDI-4 + EXT-3-Original experiments involved retraining and are not direct cross-dataset inference by the original checkpoint.

Segmentation specimen identities were inferred from filenames rather than independently verified metadata. One annotator produced the masks without inter-annotator agreement or expert boundary review, and deterministic single-label overlap handling simplified ambiguous tissue. Attention runs differed in augmentation, epoch budget, early stopping, software version, placement, and auxiliary loss; the combined extension additionally used a mixed grouped-stratified split. These results are comparative records rather than a strict one-factor ablation.

Robustness evidence covers selected synthetic conditions only, and classification and segmentation used different corruption families and reporting rules. The segmentation table contains only the five largest positive gains. XAI remains qualitative. Fourier preprocessing introduces latency and false-positive trade-offs, and mobile evidence is limited to a prototype device observation without formal farmer testing, complete delegate evaluation, or proof of final segmentation deployment. The system supports preliminary screening and is not a clinical or veterinary diagnostic method.

The revised mobile evidence improves device coverage from one bounded observation to three Android phones, but it remains an observational runtime study rather than a controlled cross-device benchmark. Workload composition differs across the three logs, the application timer is not decomposed into pure network stages, and no iOS device was available. The real-world photographs were captured with the three team cameras under varied acquisition conditions, but they do not have expert or laboratory disease ground truth; consequently they cannot support a field-accuracy claim.

17. Recommendations

Future classification work should use validation-only model selection, retain an untouched final test partition, and run at least three to five seeds for the CE and proposed configurations. Frozen prediction files should support paired analysis, classwise reports, count and normalized confusion matrices, cal-

ibration, confidence-based failure analysis, and reproducible hashes for datasets, splits, checkpoints, environments, and conversions.

The segmentation dataset should receive independent multi-annotator review with mask-IoU or boundary-agreement analysis. Specimen identities should be strengthened through source metadata and duplicate detection, while external testing should preserve specimen, farm, and source groups. A unified corruption benchmark should apply common families, severity levels, aggregation rules, and full positive-and-negative reporting to both tasks.

The Council's conditional request for a U-Net comparison was not converted into an unverified numerical row. The original/standard U-Net formulation is a semantic pixel-classification architecture and does not inherently separate multiple instances, whereas the present task is evaluated as two-class instance-segmentation with instance-level mAP and Healthy-negative diagnostics. Dice or semantic IoU from a standard U-Net cannot be directly ranked against YOLO instance mAP without a matched conversion/evaluation protocol. Instance-aware U-Net-derived variants exist in the literature, but no reproducible implementation was completed under the same grouped-specimen manifest, approximately 15M-or-lower model budget, training budget, annotation semantics, and metric definitions before the revision deadline. A fair future benchmark should use one shared manifest and report common semantic metrics (Dice/IoU/mIoU) plus an explicitly defined instance-extraction protocol if instance-level comparison is required.

The endpoint comparison with RTMDet-Ins-S is already retained in Table 6.9B. If verified RTMDet train/validation logs become available, the next revision should add a matched YOLO11n-seg versus RTMDet-Ins-S optimization-trajectory figure at this point in the segmentation discussion. Until those logs are available, no train/validation curve should be reconstructed from final metrics.

Deployment evaluation should compare quantized and FP32 artifacts under one interface contract and separate decode, preprocessing, initialization, inference, postprocessing, and UI timing. Cross-device CPU and supported-delegate profiling should record warm-up, thread count, hashes, app version, and device identifiers. The final segmentation checkpoint requires explicit TFLite/LiteRT export and an end-to-end mask regression suite. A controlled user study should verify that confidence, masks, limitations, and escalation guidance are understood without being interpreted as diagnosis.

18. Personal Reflection

The principal methodological lesson from this project is that a large metric is not sufficient evidence by itself. Several apparently simple decisions - the split policy, the source regime, which artifacts were retained, whether Healthy negatives were evaluated, and whether two timing numbers came from the same execution path - materially changed how the results could be interpreted. The specimen-grouped segmentation study was especially important because it showed that image-level random splitting could produce much larger localization scores while preserving substantial specimen overlap. That experience shifted the emphasis from maximizing a headline value to documenting why the value is credible.

A second lesson was the importance of separating model research from system evidence. A GPU screening latency cannot be presented as Android latency, and a more accurate disease-region model is not automatically a safer deployment choice if it generates many false disease masks on Healthy images. The RTMDet comparison made this trade-off concrete: stronger localization on disease images coexisted with severe Healthy-negative behavior and a larger model footprint. Likewise, the Fourier experiments showed that an apparently useful preprocessing operation could help under one augmentation regime and hurt under another. These cases reinforced the need to report the conditions under which a result holds rather than converting every improvement into a universal claim.

18. Personal Reflection (continued)

The classification extensions produced a similar lesson. The proposed configuration improved the primary SDI-4 result and the separately trained EXT-3-Original experiment, but it did not dominate the partition-wise integrated four-class regime. Rather than treating the reversal as an inconvenient outlier, the final report uses it to bound the conclusion: source composition, class priors, and retraining conditions matter. The same discipline applies to the XAI and real-world mobile observations. Activation maps are qualitative evidence rather than causal explanations, and successful execution on three Android devices demonstrates feasibility under the tested application contract rather than field diagnostic accuracy or a universal hardware ranking.

For future work, the strongest improvement would be procedural as much as architectural: freeze manifests and predictions, use validation-only model selection, preserve exact environment and conversion hashes, repeat the principal comparisons across several seeds, and keep negative controls beside positive-task metrics. This project ultimately made the team more cautious about what constitutes a defensible result. The goal is not to weaken the contribution, but to make each claim traceable to an artifact, a protocol, and an explicit interpretation boundary.

VII. Conclusion

1. Summary of Findings

1.1 Project Scope and Evidence Branches

The completed project established a connected research and implementation pipeline for shrimp-disease image analysis. Its first evidence branch addressed four-class, single-label classification in the fixed order Healthy, BG, WSSV, and WSSV_BG. Its second branch addressed disease-region instance segmentation with BG and WSSV instance masks, while Healthy images were retained as empty-label negative samples. These branches were supplemented by candidate-model screening, controlled robustness tests, qualitative explainability, leakage-aware evaluation, attention and Fourier investigations, and a native Android deployment path.

The evidence branches should not be collapsed into one performance claim. Classification measures image-level decisions, segmentation measures spatial localization and negative-image behavior, and the mobile study measures application-side runtime feasibility under a specific implementation. The final Android workflow is a conditional two-model cascade: the FP32 classifier always executes first, while the FP16 instance-segmentation model is invoked only for an above-threshold BG, WSSV, or WSSV_BG classification. Healthy and below-threshold outputs terminate after classification. The predicted class is therefore an application-control signal rather than verified field ground truth.

1.2 Classification Findings

Candidate screening covered 17 TIMM classifiers and 15 Ultralytics classification variants under the retained seed-42 partition. The completed TIMM ledger showed that ConvNeXt-Tiny-IN22K produced the strongest TIMM test Macro-F1 at 84.16% with 27.82M parameters, placing it outside the project-specific deployment preference of approximately 15M parameters or fewer. Within that preference, MobileNetV3-Large was the strongest retained TIMM row at 80.10% test Macro-F1 and 4.21M parameters. The Ultralytics screen subsequently selected YOLO26m-cls, which reached 89.02% Macro-F1 at 10.36M parameters in the retained representative screen. The timing ledgers are auxiliary screening evidence and are not interpreted as one cross-framework or Android latency benchmark.

The controlled classification reference was YOLO26m-cls trained with cross-entropy. The final selected configuration retained the same reference architecture and introduced the training-only ASL-LDAM objective together with SimAM-DCFR feature recalibration. Under the recorded fixed seed-42 protocol, accuracy increased from 89.02% to 91.33%, Macro-F1 increased from 89.02% to 91.01%, and Cohen's Kappa increased from 0.8505 to 0.881593 (85.05 to 88.16 on the displayed $\times 100$ scale). The Macro-F1 change was +1.99 percentage points. Because the comparison used one fixed seed, it is an observed improvement under that protocol rather than evidence of statistical significance.

The final result must also be interpreted with the retained artifact boundary. The primary prediction-level file, complete classwise report, and primary count and normalized confusion matrices were not retained for the 173-image test partition. Consequently, the overall improvement cannot be decomposed reliably into exact class transitions. Avoiding reconstruction from rounded aggregate values limits class-specific interpretation but preserves factual integrity.

Additional-source experiments further qualified the main finding. On EXT-3-Original, the proposed configuration reached 91.29% Macro-F1 compared with 72.30% for CE. On SDI-4 + EXT-3-Original, however, CE reached 86.51% Macro-F1 while the proposed configuration reached 82.18%, a 4.33-point disadvantage for the proposed method. Both regimes retrained their models after sourcewise splitting and partition-wise merging; they were not zero-shot inference by the original SDI-4 checkpoint. The opposite outcomes therefore indicate sensitivity to source composition, class priors, and acquisition differences rather than universal superiority.

Across the five retained aggregate classification corruptions, the proposed classifier exceeded CE in Macro-F1 and Kappa. The Macro-F1 gains were +7.32 points for impulse noise, +3.71 for Gaussian noise, +6.03 for contrast reduction, +5.66 for defocus blur, and +7.07 for low light. Absolute performance un-

der impulse noise nevertheless remained limited, so these results indicate relative resilience within controlled synthetic stress tests rather than solved severe-noise behavior or guaranteed farm generalization. Grad-CAM++, HiResCAM, and EigenCAM remained qualitative activation inspections and are not disease masks, causal explanations, or clinical localization evidence.

1.3 Instance-Segmentation Findings

The segmentation branch confirmed that split design materially changes apparent localization performance. The convention-matched 1,149-image subset contains 416 filename-derived specimen groups and supports specimen-grouped allocation, but the expanded 1,452-image inventory adds 303 unmatched-name images whose filenames do not expose compatible specimen identity. Those added images were allocated by image-level stratified splitting. The original subset is therefore group-aware, whereas the expanded inventory must not be described as fully specimen-leakage-free.

Healthy-negative diagnostics added information that mask mAP alone could not provide. Healthy was not a positive segmentation mask class, and Healthy images contained no positive disease-region annotation. Measuring whether those images produced retained disease masks exposed false-positive behavior that positive-only localization metrics would hide.

The baseline and comparator screen showed why model selection was multi-objective. RTMDet-Ins-S reached 48.30% full-test mAP50, 54.90% labeled mAP50, and 18.70% labeled mAP50-95, but it produced a 100% Healthy false-positive rate and used 10.153M parameters. The retained YOLO11n-seg reference was substantially smaller at 2.877M parameters and provided much better Healthy-negative behavior. The final CA-to-SimAM refinement at P3/P4/P5 used 2,854,718 parameters and reached 49.71% full-test mAP50, 55.07% labeled mAP50, and 19.64% labeled mAP50-95. These results support YOLO11n-seg as the more appropriate lightweight foundation for this application, not a claim that it dominates every metric or every detector family.

The broader attention-family comparison remains configuration-level evidence rather than a strict one-factor ablation because augmentation, epoch budget, early stopping, initialization, software version, placement, and auxiliary objectives were not identical. The separate CA-to-SimAM placement study more narrowly supports the retained P3/P4/P5 insertion decision, but it does not remove the protocol differences among the other attention families.

Fourier evidence also remained condition-dependent. With no augmentation and the hidden hook disabled, W1 ($\sigma = 50, \alpha = 0.10$) increased labeled mAP50 from 23.54% to 34.51%, but under the recorded strong augmentation policy it reduced labeled mAP50 from 59.51% to 57.71%. The direct paired spectrum study used 24 held-out images and 12 matched stochastic draws per image, producing 288 paired global observations. W1 alone increased high-band energy share by +0.000670, whereas the incremental W1 effect after strong augmentation was +0.000151, or approximately 22.5% of the original high-band effect. Edge-sensitive summaries retained different fractions of the W1 effect, including approximately 21.1% for Laplacian variance, 56.1% for mean gradient, and 79% for lesion-region gradient. The defensible interpretation is partial spectral redundancy under this seed-42 regime, not causal proof or multi-seed confirmation.

1.4 System and Deployment Findings

The final Android application integrated the complete conditional two-model bundle rather than a classifier-only prototype. The FP32 classification artifact accepts a [1, 224, 224, 3] input and produces four scores in the fixed order Healthy, BG, WSSV, and WSSV_BG. Disease-triggered branches invoke the FP16 YOLO11n-seg + CA-to-SimAM artifact for BG/WSSV instance masks. The same application build, the same two model artifacts, and the same configured four-thread CPU path were used across the three Android devices.

Each member recorded 21 application runs. On the vivo Y100, the classification-only/nosegmentation branch averaged approximately 5.016 s and the segmentation-triggered branch approximately 6.914 s. On the POCO M7 Pro 5G, the corresponding branch means were approximately 1.860 s and 2.641 s. The Samsung A32 log contained 19 segmentation-triggered runs averaging approximately 2.155 s and two below-threshold no-segmentation runs averaging 0.499 s; because it contained no Healthy branch and only two no-segmentation observations, that value must not be treated as a matched hardware

comparison. Branch composition differed across devices, so overall device averages are observational workload summaries rather than a strict hardware ranking.

The operational archive retained 21 usable real-world shrimp photographs collected with the three team-owned Android cameras under varied lighting, background clutter, viewpoint, orientation, and blur/softness; one excessively dark image was excluded. These photographs show that the deployed pipeline was exercised on non-dataset inputs under practical capture variation. They do not provide field-accuracy evidence because no veterinary, laboratory, or other expert disease ground truth was assigned. The recorded timing is application-side operational elapsed time unless a narrower timer scope is proven. No iOS/iPhone device was tested.

1.5 Overall Interpretation

The overall contribution lies in the combination of controlled model selection, explicit validity boundaries, and engineering traceability rather than in an unbounded performance ranking. The classification branch now includes a complete TIMM screen, a retained Ultralytics comparison, an imbalance-aware training objective, feature recalibration, corruption testing, multi-source qualification, and qualitative XAI. The segmentation branch combines leakage-aware evaluation, Healthy-negative diagnostics, a lightweight multi-objective model-selection rationale, attention placement evidence, and a direct Fourier-spectrum interaction analysis. The deployment branch connects both final artifacts through a conditional Android cascade and documents three-device operational runtime.

Together, these elements support a traceable preliminary image-screening system. They do not establish clinical validity, universal robustness, controlled cross-device superiority, or farm-ready disease recognition.

2. Contributions and Reflections on the Project

2.1 Technical Contributions

The classification contribution combined three levels of evidence. First, a broad seed-42 candidate screen compared 17 TIMM classifiers with Ultralytics classification candidates under a common dataset partition, showing that predictive quality, parameter count, and speed could not be reduced to one ranking criterion. Second, YOLO26m-cls was retained as the controlled reference within the approximately 15M-parameter deployment preference. Third, the final model integrated the custom single-label softmax ASL-LDAM objective with SimAM-DCFR while preserving a four-logit inference contract. ASL-LDAM remained training-only, while the SimAM-DCFR texture and channelgating paths introduced trainable feature-recalibration components.

For segmentation, the project added leakage-aware grouping, explicit Healthy empty-label negatives, a lightweight YOLO11n-seg foundation, a final CA-to-SimAM P3/P4/P5 refinement, RTMDet instance-segmentation comparators, and Fourier-domain diagnostics. The final segmentation model remained compact at approximately 2.855M parameters. The direct spectrum study further provided mechanism-oriented evidence explaining why W1 can become partly redundant once a strong augmentation policy already redistributes substantial high-frequency content.

2.2 Methodological Contributions

The direct classification comparison preserved one fixed split, class order, preprocessing representation, and evaluation scale. Macro-F1 was used as the primary outcome because the four-class dataset was imbalanced and accuracy could be dominated by larger classes. The report also disclosed that historical model-family screening inspected test-partition Macro-F1, preventing that ranking from being misrepresented as validation-only selection.

For segmentation, the methodology now distinguishes the fully filename-compatible grouped subset from the 303 unmatched-name additions, rather than overclaiming leakage control for the entire expanded inventory. Full-test and labeled-only localization, Healthy false-positive behavior, disease missed-image behavior, mask-count error, and HScore were reported separately so that positive-mask

localization could not hide negative-image failures. RTMDet and Ultralytics recall values were also kept within their recorded evaluation definitions rather than treated as automatically interchangeable.

Artifact-aware reporting remained a core methodological principle. Missing primary prediction files, classwise reports, and confusion matrices were not reconstructed from neighboring values. The final classifier parameter count and FP32 TFLite size are instead reported from the retained audited artifacts. The broader attention configurations were not relabeled as strict ablations, and the paired Fourier measurements were interpreted as diagnostics rather than causal proof.

2.3 Practical Contributions

The project carried the fixed four-class mapping into a deployable FP32 classifier and integrated it with the retained FP16 segmentation artifact in one Android control flow. The application records predicted class, confidence, ranked alternatives, mask output when generated, and runtime metadata while keeping local inference separate from persistence and Firebase synchronization.

Three-device testing and the three-camera operational image set strengthen the practical evidence beyond a single prototype screenshot. They demonstrate that the common application/model bundle executed across distinct Android hardware and naturally varied inputs. The evidence remains operational rather than diagnostic: no veterinary field labels were available, image identity/order was not standardized across devices, and iOS was not evaluated.

2.4 Project Reflection

Several practices materially improved the final thesis. Reconstructing and auditing the classification screen exposed the complete TIMM evidence instead of leaving candidate identities without retained metrics. Leakage analysis changed the interpretation of segmentation results by showing why convenient image-level splitting could be optimistic. Retaining Healthy negatives made false-positive behavior visible, while RTMDet comparison reinforced that a small gain in localization cannot be evaluated independently of negative-image failure and model complexity.

The project would have been stronger with a formal experiment registry from the beginning. Every run should have frozen its dataset manifest, environment, command, checkpoint hash, prediction file, evaluation report, and timing scope. Broader multi-metric selection rules, matched one-factor ablations, independent mask review, and exact final-artifact audits would have reduced the remaining uncertainty.

3. Limitations and Future Work

3.1 Classification Limitations

The primary classification result uses one fixed seed and does not provide a run-to-run distribution or statistical significance. Candidate screening inspected test-partition Macro-F1, so model-family selection did not preserve a completely untouched selection-independent test. Although the TIMM screening ledger is now complete, its timing fields are treated as within-TIMM auxiliary metadata and are not assumed to share an identical timing harness with the retained Ultralytics measurements. The final primary prediction file, classwise report, and confusion matrices were not retained, which prevents exact error-transition analysis.

EXT-3-Original and SDI-4 + EXT-3-Original were retraining experiments rather than direct external inference. They therefore measure adaptation under changed source composition, not zero-shot generalization of the original checkpoint. The final ASL-LDAM + SimAM-DCFR classifier was audited directly from the retained artifact at 10,886,467 parameters; its FP32 TFLite artifact, yolo26m_asl_ldam_simam_dcf_rfp32.tflite, is 41,443,690 bytes. These values are not borrowed from the plain YOLO26m-cls reference row.

3.2 Segmentation Limitations

Specimen identity is inferred from filenames. This supports group-aware allocation only for the 1,149 convention-matched images; the 303 unmatched-name additions cannot be assigned by compatible specimen identity, so the complete expanded split is not fully specimen-leakage-free. Masks were produced by one project annotator without independent expert review or inter-annotator agreement, and the deterministic WSSV-priority rule for overlapping project-authored polygons is an annotation convention rather than a biological statement.

The broader attention-family experiments use heterogeneous protocols and are not strict one-factor ablations. RTMDet-Ins-S provides a useful endpoint comparator, but its recall definition is not numerically identical to Ultralytics-exported recall, and its 100% Healthy false-positive rate prevents a simple mAP-only selection argument. No matched original U-Net experiment was completed. Because a standard U-Net performs semantic pixel classification rather than native instance segmentation, it is not treated as a required future comparator for the present instance-segmentation endpoint. RTMDet-Ins-S provides the retained external instance-segmentation comparison; any future U-Net-derived study would require a separately defined semantic evaluation protocol rather than direct substitution into the instance-level mAP comparison.

3.3 Robustness and Explainability Limitations

Classification and segmentation used different corruption families and reporting rules, so their robustness values cannot be pooled into one benchmark. Synthetic transformations do not reproduce every farm, camera, water condition, compression pipeline, or disease stage. The W1 spectrum study is based on the retained seed-42 sample and matched stochastic draws; it supports a mechanism-oriented interpretation of spectral overlap with augmentation but does not establish causality or multi-seed generality.

Grad-CAM++, HiResCAM, and EigenCAM remain qualitative. Heatmap agreement may reveal obvious body-focused behavior, but it does not establish causal reliance on a disease sign, and heatmaps are neither clinical evidence nor substitutes for independently reviewed segmentation masks.

3.4 Deployment Limitations

The revised deployment evidence covers three Android phones and the final conditional two-model pipeline, but it remains an observational runtime-feasibility study. The devices did not execute a standardized image-by-image workload, branch composition differed across logs, and the application timer may include decoding, preprocessing, initialization, postprocessing, mask reconstruction, and interface overhead. Overall device means therefore cannot be used as a strict hardware ranking or as isolated neural-network latency.

The real-world image set improves capture diversity but lacks veterinary or laboratory disease ground truth, so it cannot support mobile accuracy, sensitivity, specificity, disease prevalence, or clinical validity. No iOS/iPhone device was available. Formal farmer usability testing, expert-labeled field validation, thermal-stability testing, and systematic CPU/GPU/NNAPI/LiteRT delegate comparison also remain outstanding.

3.5 Prioritized Future Work

Future work should concentrate on strengthening research validity and cross-dataset evidence rather than simply expanding the number of model families. Most core reproducibility artifacts have already been preserved, so the next stage should build on that foundation through the following priorities:

1. Adopt a broader multi-metric model-selection framework. Future classification screening should not depend on one predictive score alone; Macro-F1 should be interpreted together with accuracy, classwise behavior when retained, model complexity, artifact size, robustness evidence, and deployment constraints. For segmentation, localization metrics should continue to be read together with Healthy false-positive behavior, disease missed-image behavior, and mask-count error. The

fixed seed-42 protocol remains the principal controlled setting for the current research direction; the priority is richer evaluation, not simply additional seed repetitions.

2. Extend the completed direct artifact audit of the final ASL-LDAM + SimAM-DCFR classifier. The retained best.pt has been audited at 10,886,467 parameters and the FP32 TFLite artifact at 41,443,690 bytes; future reproducibility work should bind these verified values to complete tensor metadata, converter metadata, cryptographic identity, and release-manifest provenance. This remains an artifact-verification task for the selected final configuration; it does not imply training SimAM-DCFR without ASL-LDAM or replacing the retained candidate-screen parameter ledger.
3. Strengthen segmentation validity through independent expert or multi-annotator review and more general leakage controls. SDI-4 supports specimen grouping because compatible specimen identifiers are encoded in its filenames, but this assumption does not transfer to datasets without equivalent naming conventions. Future data pipelines should therefore combine explicit acquisition identifiers where possible with cryptographic or perceptual hashes, near-duplicate detection, acquisition metadata, and other similarity-based checks to detect repeated or highly related images before partitioning.
4. Run narrower matched studies for the retained attention and Fourier questions under identical data, initialization, optimization, augmentation, software, checkpoint-selection, and evaluation rules. The purpose is to isolate the effect of the retained mechanisms rather than reopen an unrestricted architecture search. A U-Net-derived comparator is not prioritized because standard U-Net is a semantic-segmentation architecture rather than a native instance-segmentation endpoint; RTMDet-Ins-S already provides the retained external instance-segmentation comparison.
5. Progress from expert-labeled field validation to a farmer-facing pilot. The first translational step should collect prospective shrimp images with veterinary, laboratory, or suitably qualified expert disease labels and independently reviewed annotations. Only after that evidence is available should the system be evaluated in a controlled farmer pilot focused on capture workflow, result comprehension, reliability, and appropriate use of the screening output. Until this sequence is completed, CVio should remain positioned as preliminary screening support rather than a validated diagnostic system.
6. Treat platform expansion as a secondary engineering research direction. Future work may evaluate supported Android CPU, GPU, NNAPI, and LiteRT delegate configurations under clearly documented conditions and may extend the application to iOS when a compatible device and reproducible run-time path are available. Platform-specific measurements should remain bounded to the tested configuration and should not be inferred across Android and iOS environments.

3.6 Closing Perspective

The revised evidence changes the final interpretation in three important ways. First, classification model selection is now supported by a complete TIMM screening ledger rather than by candidate names alone, while the final YOLO26m-cls choice remains bounded by the project's predictive-efficiency criteria and the approximately 15M-parameter deployment preference. Second, the segmentation conclusion is no longer based only on one YOLO family comparison: RTMDet provides an independent instance-segmentation endpoint, the CA-to-SimAM placement study supports the retained final refinement, and the Fourier spectrum analysis explains why conventional augmentation can reduce the incremental effect of W1.

Third, deployment evidence now concerns the actual classifier-plus-conditional-segmentation application on three Android devices. This is stronger operational evidence than a single classifier screen, but it also makes the remaining boundary clearer: branch-dependent application time, unmatched device workloads, and unlabeled field photographs cannot be converted into a universal latency or disease-accuracy claim. The most defensible next step is therefore not to broaden claims, but to improve evidence quality through richer multi-metric evaluation, direct final-artifact auditing, independent annotation review, stronger cross-dataset leakage control, and prospective expert-labeled field validation.

The project establishes a traceable research and engineering foundation for lightweight shrimp-disease image screening. Its strongest result is not one isolated metric, but the combination of controlled classification evidence, leakage-aware segmentation, Healthy-negative evaluation, mechanism-oriented robustness analysis, and an operational two-model Android implementation with explicit limits on what the current evidence can establish.

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